

Recall and Unemployment[†]

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We document in the Survey of Income and Program Participation covering the period 1990–2013 that a surprisingly large share of workers return to their previous employer after a jobless spell, and experience very different unemployment and employment outcomes than job switchers. The probability of recall is much less procyclical and volatile than the probability of finding a new employer. We add to a quantitative, and otherwise canonical, search-and-matching model of the labor market a recall option, which can be activated freely following aggregate and job-specific productivity shocks. Recall and search effort significantly amplify the cyclical volatility of new job-finding and separation probabilities. (JEL E24, E32, J63, J64)

Unemployment is commonly understood as a state of job search, and is measured accordingly. Due to information imperfections, workers cannot immediately find the kind of employment that they desire and that the market offers somewhere. One leading interpretation of these search frictions is the extreme heterogeneity of jobs: by pay, schedule, location, task, work environment; and workers: by various types of skills, work ethics, collegiality, and so on. Therefore, it takes time and effort from both sides to identify and arrange a suitable match. If, however, a worker who separates from an employer and goes through a jobless spell eventually returns to work there, then much of this heterogeneity may be irrelevant, since employer and employee already know each other. In this paper, we show that recalls in the US labor market are a pervasive phenomenon, with a distinct cyclical pattern and significant implications for individual worker experiences and aggregate unemployment volatility.

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Using data from the *Survey of Income and Program Participation* (SIPP¹), we document that recalls of former employees in the US labor market are surprisingly common: over 40 percent of the employed workers who separate into unemployment (*EU* flow) return, after the jobless spell, to their last employer. This share of the flow into unemployment, which we will refer to as the “recall rate,” significantly exceeds the fraction of the same *EU* flow that is due to temporary layoffs (TL), namely, workers who report being laid off with a recall date or expectation.² In other words, recalls are more pervasive than TL. The reason is that, even within the group of permanently separated (PS) workers—those who lose their job with no indication of a recall, and start looking for another job—about 20 percent are eventually recalled by their last employer. The recall rate is even higher, over 50 percent, for the more “attached” job losers, who complete their unemployment spell without leaving the labor force (*EUE* spells). It is still substantial, about 30 percent, for all separated workers, including those who leave the labor force, either immediately after separation, such as retirees, or after some unsuccessful job search, i.e., discouraged workers.

To study the implications of recall for individual labor market experiences, we then restrict our attention to *EUE* spells, so that we can compare pre- and post-unemployment outcomes, with and without recall. Recalled workers were employed at their last job on average twice as long as new hires (6 versus 3 years of tenure), experience shorter unemployment duration (by over a month), switch occupation much less often upon reemployment (3 percent versus over 50 percent for job switchers), and stay with the employer significantly longer after the jobless spell. Negative unemployment duration dependence emerges mostly for those who are eventually recalled; the hazard rate of exit from unemployment to a *different* employer than the last one is only mildly declining over unemployment duration. Importantly, this feature of the data holds even when we consider all separations into unemployment (*EU* flow rather than *EUE* complete unemployment spells), including those who end up leaving the labor force. A natural interpretation of this evidence is that recalls circumvent to a large extent search-and-matching frictions, thus, they cannot be treated as the output of a matching function, which is only about new matches.

Next, we study the empirical relationship between recall and unemployment over the business cycle. In recessions, the probability that an unemployed worker is recalled drops, just like, but by much less than, the probability that he finds a new employer; therefore, the recall rate rises, and so does the share of recalls in the total number of hires. This increase was especially sharp during the Great Recession.

To summarize our empirical findings: when we exclude recalls from the exit rate from unemployment to employment, the probability that an unemployed worker finds a *new* job in the US labor market is on average much lower, more cyclical, and nearly independent of the duration of the unemployment spell, when compared both with the stylized facts from the CPS, where this distinction is not possible, and with our own SIPP sample when not excluding recalls. In this sense, employment

¹ <https://www.census.gov/sipp/>.

² In order for the worker to be classified as TL in both the SIPP and the *Current Population Survey* (CPS, <https://www.census.gov/programs-surveys/cps.html>), the worker must either have been given a date to report back to work or, if not given a date, expect to be recalled to his/her last job within six months.

reallocation in the US labor market is not as fluid as, and finding a new job is harder than, commonly thought, although on the bright side recall is pervasive and beneficial; the puzzle posed by Shimer (2005) is even deeper; and the negative duration dependence of unemployment is primarily tied to the fading likelihood of a recall.

Building on these facts, we quantify the importance of recalls for aggregate labor market fluctuations. We introduce a recall option in the search-and-matching model of the labor market à la Mortensen and Pissarides (1994). Jobs are hit by idiosyncratic and aggregate productivity shocks, which give rise to endogenous separations. Our key innovation is the assumption that, after separation, the productivity of the match keeps evolving. As long as the former employee is still unemployed and available, he can agree with his previous employer to rematch, induced by intervening changes in the aggregate and/or idiosyncratic components of match productivity. Recall is free and instantaneous for both parties. In contrast, firms that either cannot or do not want to recall a former employee must pay a cost to post a vacancy and search for a new worker. Similarly, unemployed workers must spend costly search effort to contact those vacancies and draw a new match.

After an endogenous separation, the firm can keep the idle position indefinitely open at no cost, hoping for conditions to improve and trigger a recall. If the firm wants to hire new workers, it can always create new vacancies: constant returns to scale in production ensure that recall and new job creation decisions are made independently. Thus, a separated worker does not need to be concerned about being replaced in his old job by a new hire. Conversely, a worker can only work for one employer at a time, and cannot scale up his labor supply like firms do with their labor demand. We limit the scope of recall to the last match by assuming that, when a separated worker accepts a new job, the previous match can no longer be recalled. Hence, a firm should be concerned about losing a former, “mothballed” employee to a new employer. The probability of this event, the (new-)job-finding probability, is the key equilibrium object in our model, as in the standard stochastic search-and-matching model, but here in part for a new reason: a higher job-finding probability reduces not only unemployment, but also recall opportunities. Recall is similar to on-the-job search, in the sense that the worker can search while still attached to an employer, but also different, because unemployed job search (while waiting for recall) and wage payments are mutually exclusive, therefore current wages cannot affect incentives to search for other jobs.

We assume that wages are set by Nash bargaining and analyze a simple equilibrium, where the option value of recall affects neither the probability of accepting a new job nor the wage the new job will pay. The only state variables are the exogenous productivity components. We calibrate the model parameters by a simulated method of moments, so that its steady-state equilibrium replicates cross-sectional moments computed from our microeconomic evidence. The calibrated model reproduces quantitatively all of our cross-sectional facts. The hazard rate of recalls declines with unemployment duration, as we observe in the data, due to dynamic selection: the longer a worker remains unemployed without being recalled, the more likely it is that the (unobserved and persistent) quality of his previous match has deteriorated since separation, and hence the less likely it is that a recall is forthcoming.

Finally, we introduce aggregate productivity shocks in the calibrated model, and simulate its stochastic simple equilibrium, with and without recall option and/or

search effort. The existence of a recall option greatly amplifies cyclical fluctuations in job separations and, when interacted with search effort, in the job-finding probability. Firms are more willing to lay off workers when they can recall them, but this is especially true in recessions, when these workers remain available for recall longer due to lack of alternatives. In model parlance, in recessions the match surplus from continuing production over temporary separation declines further. This surplus also determines the propensity to accept new matches; hence, its additional decline further reduces the average job-finding probability and depresses vacancy creation. In turn, a lack of new jobs, a lower propensity to accept them, and the recall option itself all discourage costly search for new jobs by workers, again further depressing vacancy creation. Moreover, recalls are much less procyclical and volatile than new hires, as in the data.

The rest of the paper is organized as follows. In Section I, we place our contribution in the context of the relevant literature. In Section II, we describe measurement issues that arise in the estimation of recall in the SIPP and present our preferred estimates. In Section III, we illustrate the empirical relationship between our measured recalls and employment and unemployment duration. In Section IV, we describe business-cycle patterns of aggregate recalls. In Section V, we lay out our search-and-matching model with recall that we analyze quantitatively in Section VI. Brief conclusions take stock of the results. An online Appendix presents additional materials for empirical evidence and quantitative exercises.

I. Related Literature

Several authors documented that recall of newly separated workers is surprisingly frequent and fast, and explored the implications for unemployment duration dependence. The literature on recall is entirely microeconomic in focus and relies on detailed samples that are limited often in scope and always in time span. To the best of our knowledge, we are the first to study recall in a large, nationally representative survey covering several decades, and to connect recall to the broader macroeconomic debate on cyclical unemployment.

Katz (1986) was the first to notice in 1981–1983 PSID data that observed negative duration dependence in unemployment is the result of a strongly declining hazard rate of exit to recall, masking the underlying upward-sloping or flat exit hazard to new jobs. Katz and Meyer (1990) take advantage of a supplemental survey of new Unemployment Insurance (UI) benefit recipients from Missouri and Pennsylvania for the period 1979–1981. The vast majority of survey participants (75 percent) said that they expected to be recalled, although only 18 percent had a definitive recall date; *ex post*, a sizable share were actually recalled.³ Katz and Meyer exploit these reported expectations in a competing hazard model to quantify their effect on the incentives to search for new jobs. They find that pre-displacement tenure predicts recall, which in turn predicts more favorable wage outcomes.⁴

³The definition in the CPS (see footnote 1) is likely to be stricter than the recall expectation measured in the data that Katz and Meyer (1990) used.

⁴This seminal work inspired a sizable literature, which is too large to survey exhaustively here. Fallick and Ryu (2007) use the same data as Katz (1986) and replicate Katz and Meyer's competing hazard exercise without the information on subjective recall expectations, but controlling for unobserved heterogeneity. A similar

Our sample is based on the 1990–2008 panels of the SIPP, which cover the entire US labor force for almost a quarter-century and three business cycles (1990–2013), not just UI benefit recipients, a single region, or a single recession episode. In comparison to this microeconomic literature, we confirm in our comprehensive sample the importance of recall, even for PS workers, and its empirical relationship with tenure and exit from unemployment, including its hazard rate (and wages; see our working paper, Fujita and Moscarini 2013). We also show, however, that the strongest relationship is with occupational mobility, and that recall also predicts subsequent attachment.

Recall plays a negligible role in the macroeconomic literature on unemployment. Bills, Chang, and Kim (2011) extend the canonical search-and-matching model to allow for heterogeneity in the reservation wage (value of leisure) across workers and study the amplification of aggregate shocks. To calibrate the separation probability, they use the SIPP, but only count permanent separations that do not result in a recall within four months, and target an average unemployment rate of 6 percent. This strategy presumably (although they do not say) excludes the contribution to unemployment of those workers who are separated and then recalled within the four-month period. We investigate whether the recall option affects the incentives of the firm and the worker to search for new matches, that is, whether recall and search interact, as suggested by the microeconomic literature, in which case the calibration strategy by Bills, Chang, and Kim (2011) would be problematic. In addition, we show that their choice of a four-month unemployment duration cutoff to define a recall leads to a significant underestimate of true recalls, because of data issues in the SIPP that we will discuss in detail.

Fernández-Blanco (2013) studies a similar model to ours, but only in steady state, and assumes commitment to contracts by firms. He analyzes the trade-off between providing workers with insurance (flat wage path) and with incentives not to search while waiting for a recall. In contrast, we introduce aggregate shocks and assume Nash bargaining to stay close to the canonical business-cycle model of a frictional labor market. We also aim to match with our model our estimated unemployment duration dependence preceding a recall. As Fernández-Blanco (2013) points out, one can interpret unemployment without active job search by workers who have a strong expectation of recall as “rest unemployment” in the language of Alvarez and Shimer (2011). Fujita (2004) extends the Mortensen and Pissarides (1994) model by introducing a fixed entry cost. The job can be mothballed in his model, as in our model. However, his model does not allow for a recall of the same worker and he only examines the cyclical implications for aggregate variables, such as job flows, unemployment, and vacancies.

On the empirical side of macroeconomic investigation, Shimer (2012) examines the “heterogeneity hypothesis” to explain the strong cyclical volatility of the overall

approach is taken by Jansson (2002) and Alba-Ramírez, Arranz, and Muñoz-Bullón (2007) for Sweden and Spain, respectively. Recalls amount to 45 percent of all completed unemployment spells in Sweden and one-third of all hires in Spain, and only recalls exhibit negative duration dependence of unemployment. Kodrzycki (2007) studies a sample of workers who suffered mass layoffs in Massachusetts in the early 1990s and were eligible for expensive retraining under the Job Training Partnership Act, thus arguably were not expected to be recalled at all. She finds that 4 percent of them were, against all odds, recalled, and did much better, even years later, than those who were not recalled. Nekoei and Weber (2015) find in Austrian administrative data that 58 percent of temporary layoffs and 19 percent of permanent separations are recalled, with an average recall rate of 35 percent.

job-finding probability of unemployed workers. That is, he asks whether this volatility is the result of composition effects in the unemployment pool, or rather whether all types of unemployed workers experience very cyclical job-finding opportunities. He finds that, among all observable worker characteristics in the CPS, the best case for the heterogeneity hypothesis can be made when breaking down the unemployed between TL and PS, as their proportions are cyclical and their relative job-finding chances are very different; but he still finds that this channel explains a small fraction of cyclical movement in the overall job-finding probability. The heterogeneity we consider is based on the type of exit from unemployment, recall versus different employer, as opposed to entry, TL versus PS.

Shimer leaves open the possibility of sizable composition effects in terms of unobservable worker characteristics. In order to investigate this hypothesis directly, one needs high-frequency longitudinal data with multiple unemployment spells to extract some sort of fixed effects. Moreover, the sample period needs to be long enough to cover at least several business cycles. The monthly CPS has too short a panel dimension to cover multiple spells, and each SIPP panel also has too short a time dimension to cover multiple business cycles. Hornstein (2012) tackles this question indirectly. He formulates a statistical model of unemployment duration dependence due to either selection by unobserved heterogeneity of individual job-finding probabilities or pure duration dependence such as skill loss or discouragement. He concludes that unobserved heterogeneity explains almost all of the negative duration dependence in the CPS and that the cyclical volatility of the job-finding probabilities of the long-term unemployed “types” is the main cause of overall unemployment volatility. In our data, the long-term unemployed are mostly those workers who are not recalled *ex post*. Thus, we put some empirical flesh on the traits that are “unobserved” in Hornstein’s approach. Ahn and Hamilton (2015) explain the cyclical volatility of the average job-finding probability through the composition of the inflow into unemployment by unobserved job-finding ability that they estimate with a dynamic unobserved component model. They find that the closest observable worker characteristic is PS status.

II. Measurement of Recall in the SIPP

A. Definitions: Labor Force Status and Job Identifiers

The SIPP is a collection of panels, each named after the year when it begins. In our analysis, we use the following eight panels: 1990, 1991, 1992, 1993, 1996, 2001, 2004, and 2008. Table A.1 in the online Appendix reports the period covered by each panel, which varies from three to five years. Each interview in a panel covers the preceding four-month period, called a “wave.” The first four panels, 1990–1993, have overlapping survey periods. The survey was redesigned in 1996 in a manner that introduced significant changes for our purposes. Thus, we sometimes distinguish between the first four and the last four panels, pre- and post-1996.

The SIPP assigns a unique numerical ID to each employer for each worker, for up to two jobs held simultaneously (EENO1, EENO2). This ID is the counter of the number of firms that the individual worked for until that point in time in the panel. For brevity, from now on we will refer to it as “job ID,” although it is important to

remember that it identifies an employment relationship and does not change when the worker is either promoted or otherwise asked to change duties by that employer. When a worker separates from an employer and, after a jobless spell, returns to the same employer, we call this event a “recall.” We do not study “second round” recalls that occur after one or more spells of employment at a different company, possibly without any non-employment in between.

To build the sample of relevant jobless spells, we adopt the following criteria. First, we focus on individuals who are assigned “longitudinal weights” by the Census Bureau. This allows us to study the history of workers who participated in the entire survey. These weights are designed to make this sample nationally representative in terms of observable worker characteristics over the panel period. We also exclude so-called type-Z imputed observations.⁵ We discuss below in more detail additional sample selection issues that can potentially impact our calculations and show that the effects are likely to be small.

The SIPP contains variables indicating the starting and ending dates of each job and weekly labor force status. For our analysis, we use a monthly panel. Specifically, we measure labor force status (employment “*E*,” non-employment “*NE*” that can be either unemployment “*U*” or out of the labor force “*OLF*”) for each individual in the second week of each month, in line with the measurement in the CPS. We identify “*E**NE*” completed spells of non-employment, of any positive number of months (a shorthand for *E**NE* . . . *NE*), where the individual experiences both a separation and an accession with a non-employment spell in between. To benchmark the frequency of recalls, we also identify spells of non-employment that either begin but do not end within the panel (*E**NE*), or are ongoing when the panel begins and end in employment (*NE**E*) within the panel. Later, we consider the cases where a worker separates into or is hired from unemployment (*U*), hence *EU*, *UE*, and *EUE* spells (again, the latter is a shorthand for *EU* . . . *UE*).

In measuring recalls, we restrict attention to jobless spells that begin with a separation in the first year (in the case of three-year panels) or the first two years (in the case of longer panels) of each panel. We adopt this sample selection to avoid right-censoring of jobless spells due to the ending of the panel, ensuring that the jobless spell could last roughly two years and still be measured by the survey.⁶ Similarly, to avoid left-censoring of spells that are ongoing at the beginning of each panel, when we benchmark recalls against all hires (rather than separations), we focus on jobless spells that end (with a hire) in the last year or last two years of each panel. We further checked the robustness of our results with respect to the different window size, i.e., including more separations (hires) that occur later (earlier) in the panel. Those results are similar and available upon request.

While the SIPP uniquely provides all the pieces of information we need, it contains two types of measurement errors that are particularly relevant to our analysis, one in

⁵ We thank Martha Stinson for suggesting this conservative procedure. The type-Z respondents are ones who answered very few questions of the survey and, therefore, have many of their responses imputed. The concern is that the type-Z respondents have spuriously higher recall rates, thus biasing the aggregate recall rate upward. Our results are actually unaffected by the inclusion of these observations. However, we believe that excluding them is a prudent practice. Dropping the type-Z observations reduces the sample size for our analysis roughly by 7 percent.

⁶ Even with this sample selection, some of non-employment spells are necessarily right-censored as a result of a long non-employment spell. We treat these cases as non-recall.

non-employment status (unemployment U versus nonparticipation OLF) before the 1996 panel, and one in job IDs following certain types of non-employment spells afterward.

The SIPP redesign in 1996 changed the definitions of labor force states, making them consistent with the monthly CPS, but not entirely comparable to pre-1996 SIPP panels. Figure A.3 in the online Appendix shows a permanent downward jump in OLF between the 1993 and 1996 panels, matched by an upward jump in U and TL. In contrast, we do not observe a similar discontinuity in the monthly CPS around its own 1994 redesign. We are not aware of any literature documenting or even suggesting measurement error of TL versus OLF status at entry into unemployment in the two major national surveys *after* their redesign. Since 1996, the definition of TL is identical in these two datasets and consistent over time. In Table A.6 in the online Appendix, we compare the TL share of the flow into unemployment in the SIPP and in the monthly CPS over the periods covered by each SIPP panel after 1996. The shares are of similar magnitude and relatively stable over time. We conclude that the problem exists in the SIPP before 1996, when many unemployed workers (on TL by the definition of the post-1996 panels) were erroneously classified as nonparticipants in the pre-1996 SIPP panels, presumably because they were neither employed nor engaged in active job search. Because no solution to this measurement issue is available, whenever we condition our analysis on labor market status, we focus on post-1996 SIPP data.

Stinson (2003) showed that job IDs in the SIPP 1990–1993 panels were subject to substantial miscoding, and then corrected the problem using confidential employer name information and administrative data containing employer-level job counts. This revision of job IDs makes it possible for us to correctly identify recalls in these early panels. We therefore view the aggregate recall rate computed from the 1990–1993 panels as reliable. This is a critical assumption on which we build our entire empirical analysis. We will provide corroborating evidence supporting this assumption.

In the mid-1990s the Census Bureau introduced computer-assisted personal interviewing (CAPI), and redesigned the SIPP in its first new panel after that date, 1996. Among other improvements, CAPI reduced the level of post-collection edits and imputation, and helped to maintain longitudinal consistency, including of job IDs, obviating the *ex post* revision that had been necessary for the 1990–1993 panels. These improvements, however, did not eliminate one issue that already existed concerning job IDs. When a worker separates from an employer and is jobless for an entire four-month wave, the SIPP, by default, assigns different job IDs to the employers before and after the jobless wave(s), even if in reality they are the same company. The one important exception is when the worker is placed on TL, in which case the SIPP carries over the information about the last job ID, even after a long unemployment spell.⁷ Stinson (2003) resolved this problem in the 1990–1993 panels by retrospectively considering information on each individual record for the entire panel, which was not available to interviewers and coders when the survey was ongoing.

⁷ Presumably, the purpose of this change was to lighten the survey collection and processing load, and the rationale was that in those cases the next employer should be new anyway, because any recall of unemployed workers who are not on TL tends to happen either quickly or not at all. However, using the pre-1996 cleaned job IDs, we show that this assumption is not totally warranted.

To recap: we regard job IDs as reliable in the pre-1996 panels, while the labor market status in the SIPP is consistent with the CPS and over time only in the post-1996 panels. We will discuss other potential sources of mismeasurement that lead to the underestimation of recall rates and propose an imputation procedure to recover the missing recalls after 1996. But we first present evidence from the raw microdata.

B. Preliminary Evidence on the Incidence of Recalls

We begin with empirical evidence on the frequency of recall among completed jobless spells $E\bar{E}E$. Table 1 contains our main findings. The first two columns report the number of completed spells and the fraction that end in recall in the raw data. We note that this recall rate is very high, especially before 1996, but also after 1996, given that we know, for reasons explained in the previous section, that it is underestimated.

Three tables in the online Appendix provide more detailed evidence on these results and on their robustness. First, Table A.2 reports the same recall rate, share of all separations that start a complete jobless spell ending in recall, in the 1990–1993 panels that we take as accurate, broken down by various worker and job characteristics. Recall is much more prevalent among older workers and union members, working in goods-producing sectors. But younger, non-unionized workers in service sectors still are recalled frequently and represent the vast majority of the US workforce. The aggregate recall rates thus reflect more closely the latter groups. Gender and education do not make much of a difference for the recall rate. Next, Table A.3 reports recall rates as a share of all non-employment spells, $E\bar{E}$ and EU , many of which are not completed, because the workers end up in retirement or persistent nonparticipation, hence are not even available for a recall. The share of all jobless spells that end in recall is only slightly lower compared with complete spells in Table 1. Finally, Table A.4 reports the fraction of *hires* that are recalls. The numbers are almost identical to those of separations: a large share of all hires are of former employees. We again observe a drop in recall rates following the 1996 redesign. Before addressing this drop, we briefly discuss two potentially important measurement issues which may bias our estimates of average recall even before 1996: selective attrition and seam bias.

Accurate job IDs in the pre-1996 panels are not sufficient to guarantee accurate measurement of recall rates. A specific concern is survey attrition. The SIPP is not address-based, like the CPS, but aims to track respondents also when they move. Nonetheless, this may not always be possible, and some correlation between geographical mobility and attrition may be unavoidable. If TL workers are more likely than PS workers to either stay put waiting for a recall, or answer the survey because they are less busy looking for work, they may be less subject to survey attrition, and this increasingly skews the unemployed sample toward workers who are more likely to be recalled.⁸ Our average recall rates use longitudinal weights to correct for differential attrition by observable worker characteristics. We then estimate a Probit regression of attrition on a rich set of demographics and on labor force status (TL,

⁸Indeed, attrition rates in the SIPP are high. Slud and Bailey (2006) estimate in the 1996 panel that 30 percent of all respondents to Wave 1 did not complete the survey. They examine implications for some variables, but not for recall. In our longer sample period we find even slightly higher attrition rates.

TABLE 1—INCIDENCE OF RECALL AMONG JOB SEPARATIONS FOLLOWED BY COMPLETE JOBLESS SPELLS

Panel	Actual		Actual + imputed						
	<i>E#E</i>		<i>E#E</i>	<i>EUE</i>		TL		PS	
	Spell count	Recall rate	Recall rate	Spell count	Recall rate	Spell count	Recall rate	Spell count	Recall rate
1990	3,325	0.371	0.371						
1991	2,310	0.423	0.423						
1992	2,827	0.407	0.407						
1993	2,587	0.398	0.398						
1996	8,341	0.190	0.319	3,384	0.449	1,481	0.846	1,903	0.172
2001	3,904	0.209	0.328	1,553	0.455	678	0.867	875	0.168
2004	3,730	0.226	0.328	1,369	0.491	663	0.865	706	0.177
2008	4,935	0.262	0.412	2,756	0.532	1,354	0.866	1,402	0.236

Note: SIPP: separations occur in waves 1–3 in the 1990–1993 and 2001 panels, in waves 1–6 otherwise.

PS, and OLF) at separation as a proxy of unobservable heterogeneity in the propensity to be recalled. We find that, although the attrition rate is significantly larger in expansions and for job losers (relative to employed workers), within job losers PS workers are only 0.5 percent more likely to leave the survey than TL at every wave (see Table A.8).

It is well known that many types of transitions, especially between labor force states, tend to be reported in the SIPP at the “seam” between two waves (see Bound, Brown, and Mathiowetz 2001 for a detailed statistical analysis). We can detect this phenomenon in all panels, even 1990–1993, because Stinson’s (2003) validation of job IDs in those panels focused on the identity of the employers, not on the *timing* of labor market transitions, which is measured with error. However, we see no reason why in those early panels the seam effect should bias the *average* recall rate and, in fact, we find that it does not. In online Appendix Table A.14, we consider “short” *E#E* spells with non-employment duration of less than or equal to two months, and we split this sample into two types: one where the entire *E#E* spell occurs within a wave and the other where it crosses the seam between waves. Recall rates before 1996, based on job IDs validated by Stinson, are essentially the same for these two samples (48 percent versus 49 percent). However, we do need to worry about post-1996 observations. Indeed, job IDs tend to change disproportionately when the non-employment spell crosses a seam: the recall rate drops from 48 percent of within-wave spells, the same as before 1996, to 32 percent when similar spells cross a seam.

To recap, in the SIPP, we find no evidence of mismeasurement of recall in the 1990–1993 panels, but we also identify two reasons why recall rates are underestimated in the post-1996 panels. First, job IDs are reset by default after an entire wave of non-employment, making it impossible to directly detect a recall. Second, recall rates are much lower when a short spell of non-employment crosses a seam, likely due to job ID miscoding. We now propose and implement a procedure to impute those “missing recalls.”

C. Imputation of Recall in Post-1996 SIPP Panels

To perform the imputation, we first split the sample into “short” and “long” spells of non-employment, lasting up to two months and three months or longer,

respectively. In each case, we use a “reference sample” to estimate a logit regression that predicts recall given observable worker and spell characteristics, such as non-employment duration, switching of occupation, and many others, and then use the estimated coefficients to perform multiple randomized imputations for each relevant spell. Tables A.11 and A.13 in the online Appendix report the specification and results of the imputation regressions.

For the short spells that begin as TL, we assume that job IDs, hence recalls, are measured accurately, whether or not these spells cross a seam. This is because the SIPP preserves the job ID for TL workers. For short spells that do not begin as TL, we assume that job IDs are accurate when the spell does not cross the seam, because the within-wave recall rate is identical to the pre-1996 benchmark (online Appendix Table A.14). For the remaining short spells that do not start as TL and then cross a seam, the strongest predictor of recall in the logit is 3-digit occupational mobility. To be conservative, when we observe such an occupational switch after crossing a seam, namely, when the worker reports two different occupations in the two consecutive interviews, we directly mark no recall. This choice follows from the observation that, among these short cross-seam spells, less than 10 percent of the occupational switchers in the pre-1996 panels are recalled (online Appendix Table A.14). This choice is conservative because crossing a seam introduces error also in occupational codes, turning stayers, who were likely to be recalled, into switchers. So our final imputed recall rates are still likely to be biased downward.

The reference sample for the imputation of recall after short *E/E* spells that do not start as TL, do not result in an occupational change, and cross the seam, is the sample of analogous short spells that do not cross the seam. Here, we only exploit the post-1996 data for the imputation regression, so that we can use a labor market status variable (PS or OLF), which is reliable after 1996. The recall rate after imputation is reported in the last row of online Appendix Table A.14. The imputation here does not make a major difference.⁹

The reference sample for the imputation of recall after long jobless spells post-1996 is the analogous sample in the 1990–1993 panels (i.e., *E/E* with three or more months of non-employment *E*). Because measurement of labor market status in the SIPP is not comparable before and after 1996, we do not use that information in the estimation. Hence, we also impute recalls for those on TL, even though their job IDs and recalls are measured accurately in the post-1996 panels. This is necessary to avoid selection by labor market status, which is obviously nonrandom and likely correlated with recalls.¹⁰ Online Appendix Table A.12 reports the results. The imputation raises the recall rate from 0.11 to 0.34, a level comparable to the one in the pre-1996 data. For TL workers, we impute a recall rate of 72 percent, which is very close to the actual one (77 percent) that we observe without error. This is an

⁹Part of the reason can be attributed to our assumption that, for short spells that do not start as TL and cross a seam, a change in occupation means a change in employer. Given this assumption, the imputation eliminates a few recalls that were in the raw data, and assigns a recall only to few occupation switchers on TL, reducing (from 2 percent to 1 percent) the recall rate of all occupational switchers.

¹⁰The key assumption is that the relationship between observable worker characteristics and probability of recall did not change over the last 25 years. While there is no direct test of this assumption, the average recall rate (as a share of total hires) in the Quarterly Workforce Indicators (QWI), assembled from error-free, administrative data, shows a pronounced countercyclical trend and a very modest downward trend in 1995–2012, just like our imputed recall rate in the SIPP. See Section IVB for details.

important result that validates our imputation procedure, given that it does not utilize explicitly *any* information on unemployment status (TL/PS). Evidently, the other spell and worker characteristics used in the imputation regression capture correctly the TL status and, thus, recall. In contrast to the case of short spells, the imputation of long spells makes a large difference in the aggregate recall rate.

To summarize, we impute recalls only after 1996 and only when the jobless spell either lasts three months or more, or lasts one or two months, begins not on TL, ends after crossing exactly one seam, and does not generate an occupational transition. Quantitatively, almost all action occurs in the former case, long spells, whose imputed recall rates are three times the observed ones. We can validate these imputed rates independently, as they are almost identical to the results from two reliable subsamples: the reference sample before 1996, and the TL subsample after 1996. The impact of the seam bias on short spells is much smaller, and the imputation only affords a modest correction.

In the online Appendix, we provide additional evidence of the validity of our imputation procedure based on another “in-sample forecast.” We discard randomly one-half of the (valid) observations in each reference sample and reimpute them; we recover the observed recall rates closely on average, with equal Type I and Type II errors of about 15 percent.

In Table 1, we present the estimated recall rates, by SIPP panel after 1996, resulting from our imputation procedure. Close to 40 percent of all completed jobless spells $E\cancel{E}$ end up in a recall. The center columns of Table 1 further restrict attention to “attached” workers, who separate into unemployment but stay in the labor force. Recall rates are now close to 50 percent of all complete unemployment spells EUE . These are strikingly large numbers. Table A.3 in the online Appendix repeats the exercise of Table 1 for all separations into non-employment $E\cancel{E}$, including permanent ones like retirement. Their recall rate is about 30 percent, and rises again over 40 percent for all separations that begin with unemployment, EU . Finally, Table A.4 in the online Appendix computes recalls as a share of all *hires* from non-employment; the results are almost identical to those for separations from Table 1 and online Appendix Table A.3. In both separation- and hiring-based measures, a visible drop in recall rates remains between pre- and post-1996 panels even after imputation, suggesting that our imputation procedure is conservative.

So far, we used the TL/PS classification merely to correctly build various types of jobless spells and to impute recalls after some short spells. This distinction is also of independent interest to draw the important distinction between *ex ante* expectations of a recall (TL), which is a traditional subject of investigation, and *ex post* recall outcomes, which we measure for the first time in a comprehensive manner. The last columns of Table 1 break down the complete unemployment spells EUE in the middle columns by detailed unemployment status, TL or PS, at the time of separation. While the vast majority of TL are recalled, as they (and we) expected, a sizable fraction of them still change employer. More interesting, close to 20 percent of PS workers, who did not expect to be recalled upon separation, are recalled nonetheless. This share is close to one-quarter in the 2008 panel. Because PS separations are much more frequent than TL, the contribution of these “unexpected recalls” of PS workers to the overall recall rate is sizable. The cross-sectional correlation between TL and recall is 0.67, high but still very far from one. As we will see shortly, TL and

recall differ even more in terms of cyclicalities. This key result reveals an important distinction between *ex ante* expectations of recall, as measured by TL status, and *ex post* outcomes, as measured by recall. It also provides additional reasons not to dismiss recall as a relic of the manufacturing-based, unionized economy of the 1970s and early 1980s. Finally, it motivates our focus on recall as a distinct phenomenon from the better-known TL, and the need to work with the SIPP as the best source of information on recalls.

III. Recall and Labor Market Experience

Having measured recall and shown that it occurs frequently, we now provide evidence that the labor market experience of recalled workers markedly differs from that of new hires. First, recalls are associated with stronger attachment to the employer, both before and after the jobless spell, so they appear to reflect some form of firm-specific knowledge. Second, recalls are widespread in the population and not overwhelmingly concentrated among few individuals. Third, recalls occur quickly, while workers who are not recalled spend much longer being unemployed. Fourth, the probability of recall starts high and sharply declines with unemployment duration; in contrast, unemployment spells that end in new hires exhibit modest duration dependence. This evidence will inform our modeling strategy. The first and third facts will motivate our assumption that recalls are free and instantaneous, while new hires are generated by a matching function, customarily used to formalize the costly and time-consuming meeting process between job vacancies and the unemployed that is due to imperfect information about match quality. The second fact will motivate our choice to model recall as the result of selection by *ex post* match heterogeneity, affecting *ex ante* homogeneous workers. The last two facts will inform our modeling of this selection process.

A. Employer Attachment and Recall

It is well known that the hazard rate of separation from a job is strongly declining in tenure. A standard rationale is that tenure with an employer measures some form of match-specific quality, due to either selection of good matches or accumulation of specific human capital. A recall, by definition, brings a worker back to the employer where he/she already has some tenure. Hence, we expect a positive correlation between tenure, both before and after separation, and recall. Indeed, Table 2 shows that workers who had longer tenure at the time of separation are more likely to be recalled.¹¹

To investigate whether a recall predicts employment duration with the same employer *after* the first completed *E#E* spell in a panel, i.e., the employment spell that begins with the “second” *E* in *E#E*, we use validated job IDs from the 1990–1993 panels. To compute the length of this second employment spell, we ignore subsequent separations that are followed by a recall within the time span of the panel, and we continue increasing tenure until either the worker moves to non-employment,

¹¹ Here our calculation focuses on *EUE* spells, but the same pattern emerges from *E#E* spells.

TABLE 2—JOB TENURE BEFORE SEPARATION AND PROBABILITY OF RECALL

Tenure/panel	1996	2001	2004	2008
< 1 year	0.350	0.342	0.403	0.439
1–3 years	0.454	0.414	0.456	0.523
≥ 3 years	0.635	0.645	0.649	0.639

Note: SIPP: based on *EUE* sample.

TABLE 3—MEAN EMPLOYMENT DURATION AFTER THE FIRST COMPLETE JOBLESS SPELL
(Months)

	Recall	New hire	Recall	New hire
	1990 panel		1991 panel	
All spells	10.7	6.8	11.6	7.1
Separation ≤ Wave 5	14.9	9.4	15.9	10.6
	1992 panel		1993 panel	
All spells	13.5	7.5	12.3	8.1
Separation ≤ Wave 5	19.0	10.4	18.1	12.1

Notes: SIPP: based on *EUE* sample. Mean duration can be right-censored by the end of the panel. Second and fourth rows consider only the cases where a transition into non-employment occurs at or before Wave 5.

or to another job, or the panel ends, which right-censors the spell. Table 3 reports the resulting average duration of the second employment spell in a panel including both completed and right-censored spells. Employment spells that begin in the first five waves of each panel are less likely to be right-censored, even more so in the 1992–1993 panels that have one more wave than 1990–1991. From this table, it is very clear that recall predicts more stability in the ensuing employment relationship. Hence, it looks as if pre-displacement tenure is not “reset,” but resumed upon recall.

In Table A.5 in the online Appendix, we compute recall rates by focusing on “stable” employment relationships. We select complete jobless spells that are bracketed on either side by at least three months of continuous employment in the same firm, instead of one month in our baseline calculations. This selection cuts the sample size in half, because it drops spells where new hires separate again within three months, and separation begets separation. Under this stricter selection, recall rates are even higher. This evidence is consistent with the fact that new employment relationships are more fragile than recalls.

B. Temporal Correlation of Recalls

Our main focus is on the share of non-employment *spells* that end in a recall. Are these spells concentrated among a relatively small number of *workers*, who “cycle” in and out of employment, or rather is the incidence of recall in the workforce widespread? We answer this question with data from the 1990–1993 SIPP panels, where job IDs are accurate. Of all workers who individually experience at least one recall in a panel, 77 percent experience only one recall and contribute 58 percent of all recall events. If we compute the recall rate using only the spells of those who are recalled at most once, so excluding altogether the “serially recalled” workers, the

TABLE 4—UNEMPLOYMENT DURATION AND RECALL (*Months*)

Panel	Overall			Recall			New hire		
	Mean	Stdev	Count	Mean	Stdev	Count	Mean	Stdev	Count
1996	2.50	2.14	3,384	2.26	1.79	1,605	2.70	2.37	1,779
2001	2.65	2.62	1,553	2.15	1.93	742	3.06	3.01	811
2004	2.48	2.35	1,369	2.09	1.75	719	2.86	2.76	650
2008	4.21	5.51	2,756	2.95	3.49	1,523	5.65	6.86	1,233

Note: SIPP: based on the EUE sample.

average recall rate drops to a still sizable 33 percent of completed non-employment spells, compared with about 40 percent for all such spells (Table 1).¹²

C. Unemployment Duration and Recall

We now turn to the association between recall and unemployment. As explained earlier, unemployment (as opposed to non-employment) is measured accurately only after 1996, so we focus on those panels. Table 4 summarizes the information about unemployment duration in the sample of completed unemployment spells (*EUE* sample) by their destination (recall or new hire). First note that recalls occur more quickly than new hires.¹³ Similarly, the standard deviation of unemployment duration is smaller for those who are eventually recalled. Average duration is clearly countercyclical for both. For recalled workers, in the 1996 and 2004 panels, which cover only expansion years, mean duration is 2.50 and 2.48 months, respectively. On the other hand, it is higher at 2.65 months in the 2001 panel, which includes a shallow recession, and at 4.21 months in the 2008 panel, which covers the Great Recession and the subsequent anemic recovery period. Interestingly, the proportional increase in average duration is twice as large for nonrecalls than for recalls. Similarly, from the standard deviations, the dispersion of unemployment duration across workers is countercyclical, especially for nonrecall hires.

Recent US experience rekindled interest in the prospects of the long-term unemployed. Figure 1 plots the discrete hazard functions, calculated nonparametrically, for exit from unemployment by duration. The sample covers all separations into unemployment (i.e., *EU* sample), including unemployment spells that are not completed before the end of the panel. Specifically, we compute the fraction of unemployed workers, at each duration (month) since they lost their last job, who exit unemployment to a recall (first row), a new employer (second row), nonparticipation (third row), and any of those (fourth row, summing the first three). The columns condition the hazard on labor market status in the first month of unemployment, in

¹²This second sample selection runs into a censoring problem, due to the end of the panel that leaves many non-employment spells incomplete and exaggerates temporal correlation. Presumably, the spells of rarely recalled workers are more likely to be censored, because the “serially recalled” workers cycle quickly in and out of employment. If we focus on recalls following separations that occur in the first 3 waves (12 months) of each panel, the share of single recalls rises to 83 percent. See Table A.16 for details.

¹³Mean unemployment duration in Table 4 (first column) is estimated very precisely. As a sample mean, its standard error is simply “SD” in the second column of each panel divided by the square root of “Count” in the third column. In all cases, the result is under 0.2 months. The same order of precision applies to mean employment duration in Table 3.

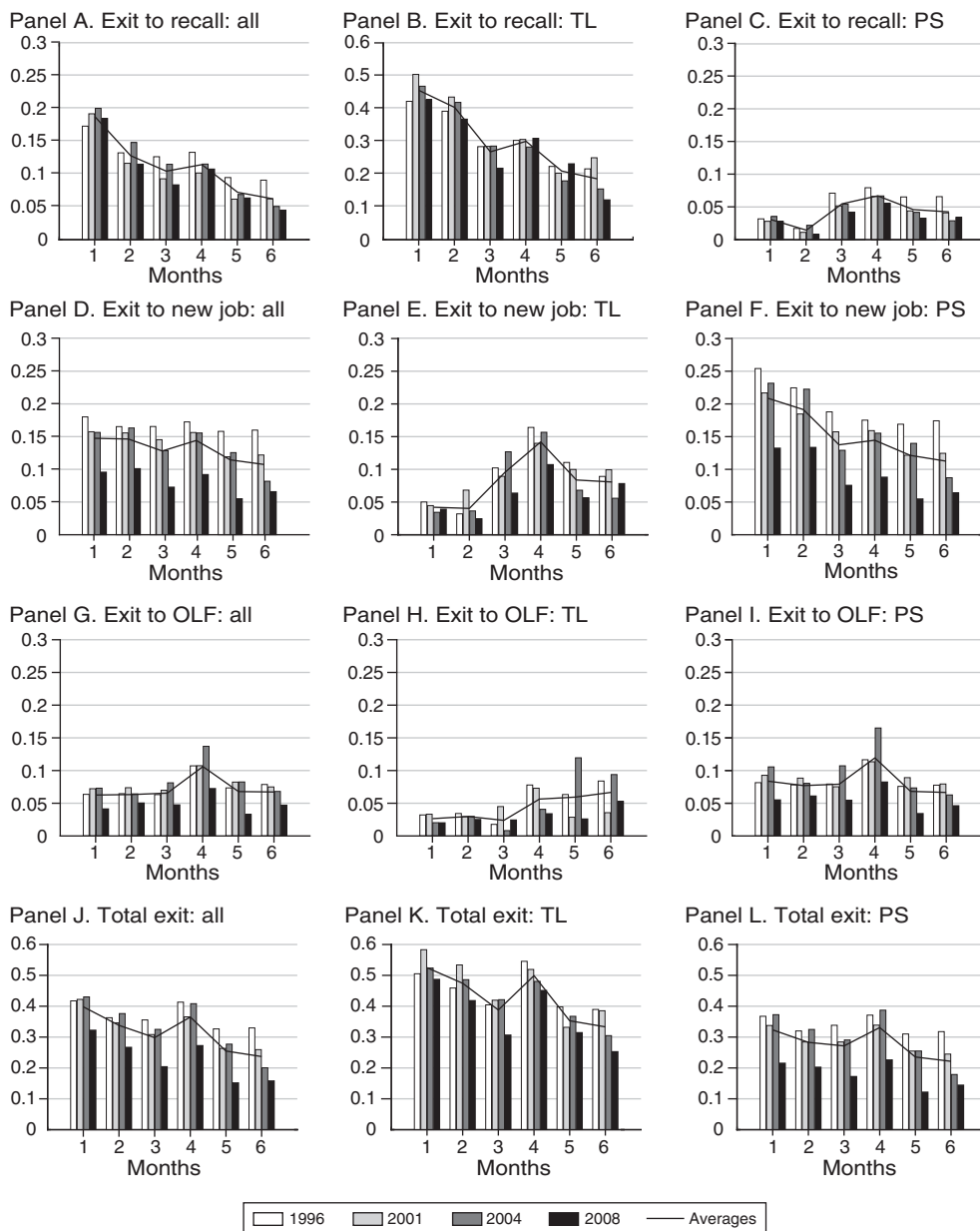


FIGURE 1. HAZARD FUNCTIONS

Notes: SIPP 1996–2008 panels: based on the *EU* sample. Labor market status (PS or TL) is based on the status at the time of separation into unemployment.

order: all and, for illustration, TL and PS. Each bar represents a different SIPP panel and the line represents the (unweighted) average across four panels. Of course, the three outcomes (new job, recall, exit to OLF) are mutually exclusive. The hazards are therefore competing, and the occurrence of an event censors the spell for other outcomes.

The first column illustrates the exit hazards from unemployment to different outcomes of job search, which add up, in the last row, to the total hazard rate. The strongest negative duration dependence appears in recalls (panel A of Figure 1), while the hazard for those who exit unemployment by finding a job at a different employer (panel D) is much flatter.¹⁴ An even flatter hazard appears for exit to nonparticipation (panel G). As a result, overall duration dependence is negative mostly due to the fading chance of a recall as unemployment continues. This is a novel and, we believe, important finding, but we stress it is limited to the first six months of unemployment, which still include the vast majority of unemployed workers.

In the second column, we examine the experience of those who begin the unemployment spell on TL. Their chance of being recalled is initially very high and sharply declines with duration (panel B). Their expectation of being recalled is clearly reflected in the next two rows: the exits to new jobs (panel E) and nonparticipation (panel H) are negligible in the first few months of unemployment and then rise when the expected recall does not materialize.

In the third column, we examine the experience of those who begin the unemployment spell with no expectation of recall (PS). In the first two months of unemployment, their chance of finding a new job is high and declines only slightly (Figure 1, panel F). After three months, that chance further drops, but the chance of a recall rises (panel C). Overall, the chance of recall is small but nonnegligible, and it appears that PS workers are recalled after they failed to secure a new job quickly. Again, exit to nonparticipation is flat in duration (panel I). Total exit from unemployment (panel L) consequently is mildly falling in duration.

Figure A.1 in the online Appendix completes the picture by illustrating the share of hires that are recalls at each duration. As should be clear from Figure 1, this share is declining in duration, but only due to the declining chance of recall of TL. Taken all together, these results suggest that negative duration dependence of unemployment is strongly related to recalls. In particular, the heterogeneity between “short-term” and “long-term unemployment types” may be directly related to the expectation/chance of being recalled or not. In turn, this chance depends on worker characteristics, but recall puts some empirical flesh on these unobserved traits.

Figure 1 also breaks down hazard rates by SIPP panel. The most salient comparison is between the 2008 panel (black), which covers a period of extremely high national unemployment, and previous periods, especially the 1996 and 2004 panels (white and dark gray) when unemployment was low. Exit rates to new jobs drop by about one-half in the 2008 panel at all durations, while exit rates to recall barely drop. This illustrates a dramatic difference in the cyclicity of the two types of reentry into employment, a finding that we will return to shortly. The well-known cyclical volatility of job-finding probabilities (Shimer 2005) is actually significantly more pronounced if we exclude recalls from accessions. Exit to nonparticipation also declines in the 2008 panel, although not nearly as much, consistent with the

¹⁴ Figure 1 in Kroft, Lange, and Notowidigdo (2013), based on monthly CPS data from 2008–2011, show that the transition rate from unemployment to employment drops by about two-thirds (one-half) when moving from zero (one) to five completed months of unemployment. Their result corresponds to the sum of panels A and D in our Figure 1, although we define months of unemployment as incomplete, hence start the hazard from one month. We observe almost identical proportional drops in the hazard rate with duration in the SIPP 2008 panel, which also covers the post-2007 period.

decline in the transition rate from unemployment to nonparticipation observed in the CPS during and after the Great Recession.¹⁵

From the last piece of evidence, it appears that recalls stabilize cyclical fluctuations in the overall job-finding probability for TL and PS workers alike and that the probability of finding *new* jobs is not only lower but also even more cyclical than previously thought. To complete our empirical investigation, we now move to explore systematically the relationship between recall and business cycles.

IV. Aggregate Time Series Evidence on Recalls

A. Survey of Income and Program Participation

The recent debate on unemployment fluctuations revolves around job-finding rates. To study how recalls impact the behavior of the overall job-finding rate, this section considers the recall rate defined as the share of *hires* that are recalls. On average, this share roughly matches that of separations that end in recall (compare Table 1 for separations with online Appendix Table A.4 for hires). We now study its cyclical properties. This evidence will inform our theoretical analysis. After dropping the observations from the first year of each panel to avoid the left censoring of *E/E* spells, we end up with 69 quarterly observations of the hire recall rate, spanning 1990:IV to 2013:II. Only since 1997:I, hence for 49 observations, we can also rely on the distinction between unemployment *U* and OLF and calculate recall shares of hires from *U*. Panel A of Figure 2 illustrates the resulting time series, seasonally adjusted by regression on seasonal dummies, logged and filtered with a cubic time trend, because gaps in the time series make HP-filtering infeasible, along with the seasonally adjusted unemployment rate from the BLS, also logged, and HP-filtered with parameter 10^5 as in Shimer (2005) (and also other series discussed below). It is possible to detect visually a rise in the recall rate during recession times, after 2001 and especially during the Great Recession of 2008–2009, as well as a sharp decline during the tight labor market of the mid-2000s. As just discussed, this is all the result of procyclical probabilities of finding employment, either at the previous or at a new employer, with the latter being much more volatile. Because the probability of finding a new job is very procyclical and that of leaving the labor force is small, the competing hazard of being recalled is countercyclical, conditional on exiting unemployment.

We supplement this fragmented time series evidence with additional evidence from the Quarterly Workforce Indicators and from the monthly CPS, which allow us to construct uninterrupted time series. While both sources are dominated by the SIPP to measure recall, they do contain useful ancillary information to understand its cyclicality.

¹⁵ Note that the decline in the labor force participation rate observed in and after the Great Recession is not inconsistent with the lower transition rate from unemployment to nonparticipation, because the transition rate from nonparticipation to employment declined even more.

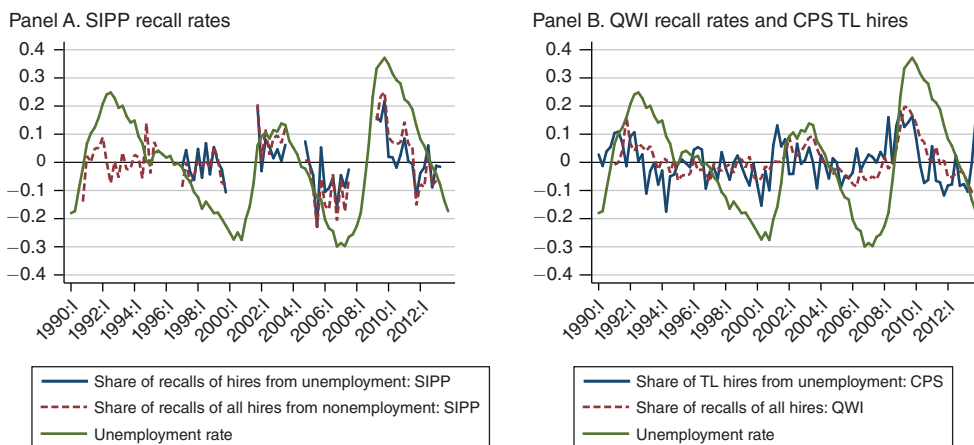


FIGURE 2. CYCLICALITY OF RECALL AND UNEMPLOYMENT RATES

Notes: All series are seasonally adjusted. The unemployment rate, QWI recall rate, and the share of TL hires from unemployment are logged and detrended by the HP filter with smoothing parameter of 10⁵. SIPP recall rates are logged and detrended by a cubic polynomial trend. The QWI recall rate is the average across the US states where the recall series are available at each point in time.

B. Quarterly Workforce Indicators

The Longitudinal Employer-Household Dynamics (LEHD) program at the Census Bureau provides a matched employer-employee administrative dataset of quarterly employment and earnings for virtually the entire US private and state sector. The LEHD has a limited time span, as states joined the program only gradually, starting in the early 1990s with several states such as California, Idaho, Maryland, Oregon, Washington, and Wisconsin. Other states joined later, many in the 2000s, and now nearly all states are in the program. The Census Bureau publishes aggregate tabulations of major labor market variables from the LEHD under the name Quarterly Workforce Indicators (QWI): see Abowd et al. (2009) for a detailed description of LEHD and QWI. One such QWI tabulation is “Recall,” the probability that a hire by an employer in quarter t had earnings from the same employer in any of the three quarters $t - 2$, $t - 3$, or $t - 4$ (but not in quarter $t - 1$, because the worker is “hired” in t). Calculated as a share of total gross hires, the QWI recall rate most closely corresponds to the results in our Table A.4. We study its level and cyclicalities.

The QWI recall rate averages about 17 percent of all hires, less than one-half of our estimate from the SIPP. Measurement of recalls in QWI differs from the SIPP in two important respects, which contribute in opposite ways to the average recall rate. Both issues arise because the underlying LEHD dataset lacks information about labor market spells. On the one hand, QWI do not distinguish between hires from non-employment and from other firms. So recalls in QWI include those that occur after the worker spent a few months with another employer. In this sense, it is a broader notion, and the recall rate should be higher in QWI than in the SIPP, where we focus on recalls from non-employment. On the other hand, QWI suffer from severe time-aggregation bias, because of their quarterly measurement. Specifically, QWI fail to detect altogether any non-employment spell that starts and

completes with a recall within a full calendar quarter. Because recalls are quick and follow mostly short non-employment spells, many of them are missed. Applying the LEHD-QWI sampling procedure to our SIPP data reduces the estimated recall rate to a level consistent with that in the QWI, providing further support to the accuracy of our measurement of recalls. Details are in the online Appendix.

The QWI recall rate is strongly countercyclical. We collect an unbalanced state panel of quarterly recall rates and unemployment rates for the 32 US states where the QWI recall series are available at least since 1999. We seasonally adjust the series, take log and HP-filter both state-level recall and unemployment rates, with smoothing parameter 10^5 . We use these state-level data in the regression analysis in the next section. In panel B of Figure 2, we present, as a summary aggregate measure, the time series of the unweighted average recall rate from these states. Its correlation coefficient with the national unemployment rate is 0.74.

C. Monthly Current Population Survey

The monthly CPS data are the most widely used source of information for labor market flows. Unlike the SIPP and the QWI, they do not contain employer information, and thus they cannot be used to measure recall. They can, however, provide a long unbroken time series of the share of hires who were on TL, out of all hires from unemployment.¹⁶ We plot this series (quarterly averages of monthly shares) in panel B of Figure 2, logged and HP-filtered with parameter 10^5 . While countercyclical, like the SIPP recall rate, its correlation with the latter is only 0.29, positive but far from perfect, highlighting once again a significant difference between ex ante TL and ex post recall.

The time span of the monthly CPS offers a longer perspective on the issue of recalls and TL. Labor market researchers paid decreasing attention to TL, due to the observed decline in its level and cyclicity (Groshen and Potter 2003) which tracked the decline in the relative importance of the manufacturing sector, where TL and recalls were common (up to 70 percent recall rate in 1965–1976: see Lilien 1980). Our empirical evidence should lead us to rethink this assessment for two reasons.

First, the decreasing incidence of TL in the CPS is observed in the stock of unemployment, but not much in the flows. Figure 3 plots unemployment stocks by reason, all expressed as a fraction of the labor force, so the sum of these three lines equals the official unemployment rate. One can see that unemployment due to TL is indeed a relatively small share of the unemployment stock, especially after the mid-1980s. Moreover, the increase in the TL stock during the last three recessions has been modest. But TL are still a much larger fraction of the flows in and out of unemployment than of the stock of unemployment. The reason for the stock-flow discrepancy is that TL spend much less time in the unemployment pool than average. So, if one is interested in worker flows, TL still matter, even today. Figure 4 shows quarterly averages of monthly probabilities of entry into and exit out of unemployment, by type of inflow, TL and PS. To construct these series, we use the duration data as in

¹⁶The *UE* flows are based on the matched records. Hires associated with TL can be identified by using the reason-for-unemployment variable.

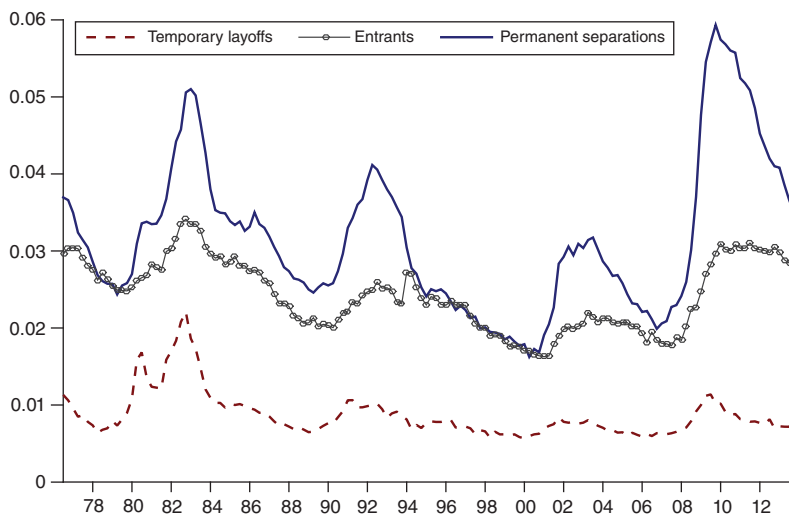
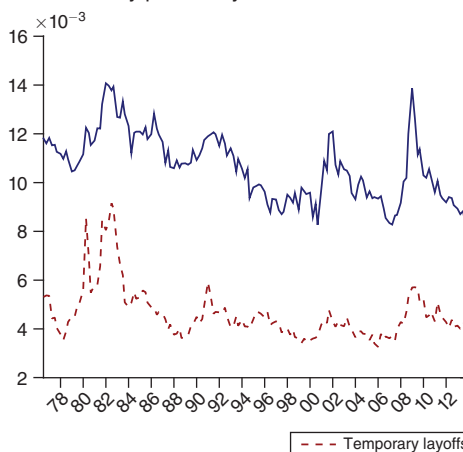


FIGURE 3. UNEMPLOYMENT STOCKS BY REASON

Note: Monthly CPS: expressed as a fraction of the total labor force.

Panel A. Entry probability



Panel B. Exit probability

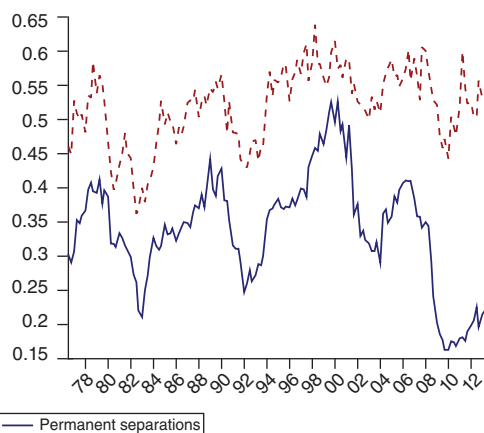


FIGURE 4. UNEMPLOYMENT ENTRY AND EXIT BY REASON

Notes: Monthly CPS: data on unemployment duration and reason for unemployment. See footnote 16 for more details.

Shimer (2012), combined with the data on reason for unemployment.¹⁷ In panel A of Figure 4, the TL inflow amounts to slightly less than one-half of the PS inflow, a ratio that has remained fairly stable since 1976. The two inflow rates move more or

¹⁷The entry probability is computed as short-term unemployment (less than 5 weeks) of each type normalized by the total employment stock; the exit probability is the outflow of each type normalized by the corresponding unemployment stock (PS or TL). By construction, exit probabilities include not only those who find a job but also those who drop out of the labor force. Due to the redesign of the CPS in 1994, the raw data exhibit a break in these series at the start of 1994. We adjust the break, following the adjustment procedure proposed by Elsby, Michaels, and Solon (2009).

less in parallel over business cycles, both showing a marked countercyclical pattern. In panel B, workers on TL enjoy a much higher exit probability than PS workers; note also that both exit probabilities exhibit the familiar procyclicality, but which is more pronounced for PS workers. After the Great Recession, the exit probability recovered rather quickly for TL, but slowly for PS.

Second, and more important, TL are only part of the story. We showed in the SIPP that PS workers, who have no clear expectation of recall, nonetheless return to their former employer with surprisingly high frequency, about 20 percent, and this frequency has not declined over the last two decades. Although this frequency of PS recall is still much lower than that of TL recall, a significant share of recalls originate from the (much larger) stock of PS workers, who did not expect a recall. Therefore, even though one can see from panel A of Figure 4 that the relative importance of TL in the *EU* inflow has diminished in the last three recessions, “recallable” workers remain quantitatively very important. Furthermore, as we will document shortly, the job-finding probabilities of unemployed workers who started the spell as TL and PS are both cyclical, while the exit rate from unemployment to recalls is much less cyclical than that to new jobs.

Figure B.1 in the online Appendix provides supplementary evidence from the monthly CPS matched records, which allow also to distinguish between exit from unemployment to employment, as opposed to nonparticipation. The PS flow into unemployment is twice as large as the TL flow, and both transition probabilities into unemployment are countercyclical. The job-finding probability of unemployed on TL is much higher but much less cyclical than for PS and entrants.

D. Summary: Business-Cycle Moments

In Table 5, we present the volatility of the various detrended recall rate measures, as well as their elasticity (regression coefficient) with respect to unemployment rates, which is our preferred measure of cyclicity. In the first four columns, we report the results based on our recall measures from the SIPP. For the QWI recall rate (fifth column), the volatility refers to the unweighted average series described above, while the elasticity is from a panel regression where we regress state-level recall rates on state fixed effects and unemployment rates. In the last column, we consider the CPS-based TL share of hires. Interestingly, all measures are similarly and highly volatile, only slightly less than the unemployment rate, as can be inferred from Figure 2. The SIPP measures are more volatile than QWI and CPS-based measures. In contrast, the unemployment elasticities are high and similar in the SIPP and QWI, which measure genuine recalls, but much smaller for the CPS share of TL accessions. This is one of our central findings: the distinction between TL and recalls is quantitatively important in terms not only of average levels but especially of their volatility and cyclicity. In short, the incidence of recalls is higher and much more countercyclical than that of TL.

V. A Stochastic Search Model with Recall

In order to make sense of this evidence and understand its relevance to unemployment dynamics, we introduce a recall option in the Mortensen and Pissarides

TABLE 5—CYCLICALITY OF RECALL RATES

	Recall rate					Share of TL Hires (CPS)
	$\#E$	UE	$E\#E$	EUE	QWI	
Volatility	0.097	0.084	0.105	0.082	0.062	0.074
Elasticity with respect to unemployment	0.348 (0.051)	0.244 (0.052)	0.412 (0.055)	0.222 (0.052)	0.246 (0.012)	0.083 (0.047)
R^2	0.406	0.316	0.390	0.316	0.162	0.042
Observations	69	49	69	49	2,317	99

Notes: SIPP, CPS, and QWI: all series are seasonally adjusted, logged, and detrended. The QWI result is from the state-level fixed effect regression described in the text. The remaining regressions use the time series of the aggregate recall rate and the national unemployment rate. Robust standard errors are in parentheses.

(1994) economy and study quantitatively its equilibrium response to aggregate productivity shocks.

A. Setup

Time is continuous. All agents are risk neutral and discount payoffs at rate $r > 0$. Firms produce a homogeneous consumption good using a CRS technology and sell it in a competitive market. The flow output from each firm-worker match equals $p\varepsilon$, where $p > 0$ is an aggregate component common to all firms, while ε is an idiosyncratic component. Both p and ε evolve according to a Markov chain: at Poisson rate λ_p a new draw of aggregate productivity p' is taken from $dP(p'|p)$, and at Poisson rate λ_ε a new match value ε' is drawn from $dG(\varepsilon'|\varepsilon)$ while the worker is employed. Here we introduce our main modeling innovation, which gives rise to a recall option: worker and firm can suspend production and, as long as the worker does not take another job, the value ε of the (potential re-)match between the employer and the worker continues to evolve, according to the same Poisson rate of arrival λ_ε and a conditional distribution $dH(\varepsilon'|\varepsilon)$, possibly different from $dG(\varepsilon'|\varepsilon)$. The lowest possible match quality is equal to zero and an absorbing state, so when ε drops to $\varepsilon' = 0$ the match becomes permanently infeasible, as it will produce nothing thereafter. Exogenous separations may be thought of as transitions to $\varepsilon = 0$. In contrast, the rest of P , G , and H are irreducible.

There are search frictions in the labor market. In order to create new matches, unemployed workers spend search effort $s \geq 0$ at cost $c(s)$ to find, at rate $s\phi$, open vacancies, which are posted at a flow cost $\kappa > 0$ as in the standard model. Cost $c(\cdot)$ is twice continuously differentiable, increasing, and convex, with $c(0) = c'(0) = 0$. Old matches that separated can be reassembled at any time at no cost to either party, if still unmatched. Let u denote unemployment (rate) and $\bar{s}$ the average search effort of the unemployed, so that $\bar{s}u$ is aggregate search effort by unemployed workers. Let $\theta = v/(\bar{s}u)$ denote labor market tightness, the ratio of open vacancies to aggregate search effort. We assume that the flow of new contacts between open vacancies and job searchers equals $m(v, \bar{s}u)$, where m is a standard continuous and homothetic matching function. Thus, by random matching each open vacancy is contacted by a searching worker at rate $q(\theta) = m/v$ where q is continuous, decreasing, and convex and $\phi = \phi(\theta) = \theta q(\theta)$ is the worker contact rate per unit of search effort.

When an unemployed worker and vacant firm do meet for the first time, they draw from a distribution F an initial match quality $\tilde{\varepsilon}$. If they accept the match and start

producing, the worker must forfeit the recall option with his former employer(s), and simultaneously acquires a job and a future recall option with this new employer. Similarly, a vacant job that holds a match of quality ε with its former employee (where $\varepsilon = 0$ if either the former employee took another job or the separation was irreversible) can either (i) wait and do nothing (“mothball” the vacancy); (ii) recall the last employee if still unemployed; or (iii) pay κ and repost the vacancy, to contact at rate $q(\theta)$ a random unemployed worker who is searching and draw from F a new match productivity $\tilde{\varepsilon}$. Free entry in vacancy creation drives to zero the expected value to a firm of searching for a new employee.

Wages in ongoing matches are set by generalized Nash bargaining, with the worker receiving a share $\beta \in (0, 1)$ of match surplus. Unlike in the standard model, the outside options when bargaining are not obvious, as now separation is not irreversible, hence not a credible threat in a noncooperative foundation. We assume that the outside option is temporary separation, until the next productivity shock occurs and triggers a possible recall; in the meantime, parties can look for other partners and better matches. Firms have no commitment power, not even to once-and-for-all lump-sum transfer, and wages are continuously renegotiated, so that search effort by either side is not contractible.

In order to match, a firm and worker who meet and draw match quality $\tilde{\varepsilon} \sim F$ gain a positive surplus not only over the alternative of rejecting the new match and continuing search, but also over waiting for $\tilde{\varepsilon}$ to improve through the law of motion H . That is, the new match quality $\tilde{\varepsilon}$ must be good enough to begin production right away; otherwise it is lost. This assumption of “No Mothballing Before Production” captures the idea that a new match requires some initial phase of discovery and experimentation through production. Therefore, an unemployed worker and a vacancy-posting firm that have just met for the first time cannot just “keep in touch.”

Our model nests the standard Mortensen and Pissarides (1994) model as a special case with no recall option ($dH(\varepsilon'|\varepsilon) = 0$), costless unemployed job search ($c(s) = 0$ and s is normalized to 1), and a degenerate distribution of new matches (F is a mass point at the upper bound of the support of G).

We restrict attention to an equilibrium where value and policy functions are defined on a very simple state domain: aggregate productivity p and, for each match, the quality ε of the current or (if unemployed) last match. In this simple equilibrium, idle workers who are searching will be willing to accept new job offers independently of their value of recall in hand. Thus, firms posting new vacancies will not need to keep track of the evolving distribution of recall values held by the unemployed, which is then not a state variable. Bellman values are time-independent functions of p and ε only. Labor market tightness θ is a function of p only. We assume that equilibrium has these properties, and then verify that the guess is consistent with all equilibrium restrictions. These properties will make equilibrium characterization and computation very tractable.

B. Match Acceptance, Separation, and Mothballing

Let $U(p, \varepsilon)$ denote the worker’s value of unemployment, where $p\varepsilon$ is the productivity of the last match, if any (otherwise $\varepsilon = p\varepsilon = 0$), $W(p, \varepsilon)$ the worker’s value of employment, $V(p, \varepsilon)$ the value of a vacant job, where $p\varepsilon$ is the current (potential)

productivity of the last employee, if any (otherwise $\varepsilon = p\varepsilon = 0$), $J(p, \varepsilon)$ the value of a filled job, $w(p, \varepsilon)$ the wage.

Since $\varepsilon = 0$ is an absorbing state and that match will never be recalled, $V(p, 0)$ equals the value of an unattached, brand new vacancy, which is zero by free entry, i.e., $V(p, 0) = 0$. Next, we examine the decision whether to dissolve or mothball the match. Neither the worker nor the firm has any incentives to give up a recall option, unless match quality drops to zero, because waiting entails no explicit or opportunity costs either to the firm, by free entry, or to the worker, who can search for other jobs whether or not he has a recall option in hand. They decide by mutual consent to mothball the match when they are both indifferent: $J(p, \varepsilon) = V(p, \varepsilon) \Leftrightarrow W(p, \varepsilon) = U(p, \varepsilon)$. Except for a transition to the absorbing state $\varepsilon' = 0$, separation is never irreversible, but always results initially in mothballing. In this notation, we can study the decision to accept a new match and formalize our “No Mothballing Before Production” assumption as follows: *for new matches $\tilde{\varepsilon} \sim F$ only,*

$$J(p, \tilde{\varepsilon}) \leq V(p, \tilde{\varepsilon}) \Rightarrow J(p, \tilde{\varepsilon}) = 0.$$

Next, we examine the decision whether to accept or reject a new match. Search effort and the recall option give rise to moral hazard. The old employer could offer the former employee a flow payment, a firm-sponsored unemployment insurance, to discourage search for new jobs or to compensate the worker for rejecting any new offer. In turn, new employers could promise a higher wage to respond to the old employer’s counteroffer. We rule out any such competition because it would require commitment. The worker, anticipating this, will simply compare the values that he would obtain by bargaining independently with either the current or the new employer. Similarly, the last employee of a currently vacant job may want to compete with any new hire prospect (in order to keep his old job available) and retain his recall option. This competition is ruled out by constant returns to scale in production and free entry, because the firm can always create a new job for the new applicant and keep the old job “mothballed” for the former employee.

Therefore, after meeting and jointly drawing an initial match quality $\tilde{\varepsilon} \sim F$, the firm and the worker, who carries a quality ε from his last mothballed match with another firm, create the new match if and only if this yields (i) the firm more than both giving up the new vacancy (which has zero value by free entry) and mothballing the new match immediately ($J(p, \tilde{\varepsilon}) > V(p, \tilde{\varepsilon})$), and (ii) the worker more than continuing the job search, either with no match in hand or waiting for a recall of the old match ε , and also more than mothballing the new match immediately: $W(p, \tilde{\varepsilon}) \geq \max\{U(p, 0), U(p, \varepsilon), U(p, \tilde{\varepsilon})\}$. Clearly, $U(p, \tilde{\varepsilon}) \geq U(p, 0)$ for all $\tilde{\varepsilon} \geq 0$, because the worker can always reject recall of old matches and mimic a worker who has no recall option. Similarly, $V(p, \tilde{\varepsilon}) \geq V(p, 0) = 0$ for the firm.

To recap, a new match $\tilde{\varepsilon}$ will be acceptable if and only if

$$J(p, \tilde{\varepsilon}) \geq V(p, \tilde{\varepsilon}) \quad \text{and} \quad W(p, \tilde{\varepsilon}) \geq \max\{U(p, \varepsilon), U(p, \tilde{\varepsilon})\},$$

i.e., if it yields both parties a positive surplus from forming the new match and producing output immediately, and also yields the worker a positive surplus over (the option to recall) the old match, whose quality evolved to the current value ε . By the

private efficiency of Nash bargaining, $J(p, \tilde{\varepsilon}) \geq V(p, \tilde{\varepsilon}) \Leftrightarrow W(p, \tilde{\varepsilon}) \geq U(p, \tilde{\varepsilon})$. Hence, denoting by $I\{\cdot\}$ the indicator function, the worker incentive constraint is weakly more binding, and the probability that a new contact results in an acceptable match equals

$$a(p, \varepsilon) = \int I\{W(p, \tilde{\varepsilon}) \geq \max\langle U(p, \varepsilon), U(p, \tilde{\varepsilon}) \rangle\} dF(\tilde{\varepsilon}).$$

C. Bellman Equations

We can now write the (Hamilton-Jacobi-)Bellman equations that these values solve. For the employed worker, it is written as

$$(1) \quad rW(p, \varepsilon) = w(p, \varepsilon) + \lambda_p \int [\max\langle W(p', \varepsilon), U(p', \varepsilon) \rangle - W(p, \varepsilon)] dP(p'|p) \\ + \lambda_\varepsilon \int [\max\langle W(p, \varepsilon'), U(p, \varepsilon') \rangle - W(p, \varepsilon)] dG(\varepsilon'|\varepsilon).$$

Endogenous separation may follow either aggregate (p') or idiosyncratic (ε') shocks.

A worker may be unemployed and searching for one of three reasons: (i) the match was hit by an exogenous destruction shock, which sets $\varepsilon = 0$ and voids any recall possibility; (ii) the match was mothballed following a productivity shock, either aggregate or idiosyncratic, but might still be recalled; and (iii) off the equilibrium path, the firm and the worker disagree on the wage and, as a threat point, suspend production and search, until the next shock hits. Whatever the reason, an unemployed worker who currently holds an old match value ε and contacts an open vacancy expects a capital gain equal to

$$\Omega(p, \varepsilon) := \int I\{W(p, \tilde{\varepsilon}) \geq U(p, \tilde{\varepsilon})\} [W(p, \tilde{\varepsilon}) - U(p, \varepsilon)] dF(\tilde{\varepsilon}).$$

Under our assumptions, the search problem of the unemployed worker has a unique solution

$$s^*(p, \varepsilon) = \arg \max_{s \geq 0} \{s\phi(\theta(p))\Omega(p, \varepsilon) - c(s)\},$$

so that the value of unemployment solves

$$(2) \quad rU(p, \varepsilon) = b + \lambda_p \int [\max\langle W(p', \varepsilon), U(p', \varepsilon) \rangle - U(p, \varepsilon)] dP(p'|p) \\ + \lambda_\varepsilon \int [\max\langle W(p, \varepsilon'), U(p, \varepsilon') \rangle - U(p, \varepsilon)] dH(\varepsilon'|\varepsilon) \\ + s^*(p, \varepsilon)\phi(\theta(p))\Omega(p, \varepsilon) - c(s^*(p, \varepsilon)).$$

In words, after each shock to either p or ε , the worker may propose to reactivate the old job, and at all times, he can search for a new job.

We now move on to the firm, starting with the value of a filled job. The flow return equals flow output, minus the wage, plus capital gains or losses after each type of shock, which may induce the match to separate:

$$(3) \quad rJ(p, \varepsilon) = p\varepsilon - w(p, \varepsilon) + \lambda_p \int \left[\max \langle J(p', \varepsilon), V(p', \varepsilon) \rangle - J(p, \varepsilon) \right] dP(p'|p) \\ + \lambda_\varepsilon \int \left[\max \langle J(p, \varepsilon'), V(p, \varepsilon') \rangle - J(p, \varepsilon) \right] dG(\varepsilon'|\varepsilon).$$

The value of a vacant job solves a more complex equation:

$$(4) \quad rV(p, \varepsilon) = \lambda_p \int \left[\max \langle J(p', \varepsilon), V(p', \varepsilon) \rangle - V(p, \varepsilon) \right] dP(p'|p) \\ + \lambda_\varepsilon \int \left[\max \langle J(p, \varepsilon'), V(p, \varepsilon') \rangle - V(p, \varepsilon) \right] dH(\varepsilon'|\varepsilon) \\ - s^*(p, \varepsilon) \phi(\theta(p)) a(p, \varepsilon) V(p, \varepsilon).$$

The firm can offer to recall the former employee after any shock, but can also lose the recall option and be left with an unattached vacancy which, by free entry, is worth zero. This occurs (third line of (4)) if the former employee contacts another open vacancy (at rate $s^*(p, \varepsilon) \phi(\theta(p))$) and draws a new acceptable match, which has a chance equal to $a(p, \varepsilon)$. The firm could also pay the flow vacancy cost κ to meet a new worker and hire him if the new match draw $\tilde{\varepsilon}$ guarantees a positive surplus and a higher value to the firm than the continuation value of waiting for a recall. Again, the net value of this option is zero by free entry and does not appear in (4). Therefore, $V(p, \varepsilon)$ measures only the value of the recall option for the firm, while the corresponding value for the worker $U(p, \varepsilon)$ also contains an option value of searching for another job in addition to the value of leisure.

D. Free Entry Condition and Equilibrium State Space

Firms post new vacancies until their net value is zero: for all p , $V(p, 0) = 0$. After matching, if the quality ever drops to $\varepsilon = 0$, an absorbing state, the match will never be productive again and the vacancy becomes worthless, just like new ones: $J(p, 0) = V(p, 0) = 0$.

Free entry thus implies that the vacancy posting cost κ equals the contact rate $q(\theta(p))$ times the expected surplus from a contact. Here is where the assumption on the state space has bite. If the incentives of a job applicant to accept a new match draw depend on the value of the recall option that he holds, then both the probability that a vacancy is filled and the profits from filling it, hence the free entry condition, will depend on the distribution of recall values (or equivalently the qualities of last jobs held) among unemployed workers. This is an infinitely dimensional object, which evolves stochastically with aggregate productivity p .

To prove that agents can ignore this state variable and confirm the guess of a simple state space (p, ε) , we have to show that *on the equilibrium path*, where a worker is unemployed only when his last match has negative surplus, but not due to bargaining disagreement, both the probability $a(p, \varepsilon)$ that a new match is acceptable and the profits $J(p, \tilde{\varepsilon})$ that the firm earns from it are independent of the value $U(p, \varepsilon)$ of the recall option that the worker may currently have in hand. By Nash bargaining, this also requires that the worker's continuation value $W(p, \tilde{\varepsilon})$ from the match, hence his new wage, be independent of $U(p, \varepsilon)$.

The argument for the continuation value $W(p, \tilde{\varepsilon})$ follows from the lack of commitment and ex post competitions for a worker between firms. The new employer bargains with all workers in the same way, as if they had nothing in hand and offers all new hires a value $W(p, \tilde{\varepsilon})$. This is accepted if and only if $W(p, \tilde{\varepsilon}) \geq U(p, \tilde{\varepsilon})$, by the assumption of "No Mothballing Before Production." This value $W(p, \tilde{\varepsilon})$ is independent of the current recall option encoded in ε .

The argument for the probability of accepting a new match $a(p, \varepsilon)$ to be independent of the value of the recall option in hand $U(p, \varepsilon)$, thus of ε , is more subtle, and by revealed preferences. If the old match ε and new match $\tilde{\varepsilon}$ satisfy $U(p, \tilde{\varepsilon}) < W(p, \tilde{\varepsilon}) < U(p, \varepsilon)$, the worker will continue waiting for a recall, although the new match $\tilde{\varepsilon}$ would be acceptable absent the recall option. The critical observation is that any new match that is acceptable to an unemployed worker who does not hold a recall option is also acceptable to an unemployed worker who does. If the worker who makes contact with a new vacancy is jobless, his recall value $U(p, \varepsilon)$ must be low enough not to justify recall of the previous match; otherwise he would have recalled the match and not be jobless and searching, and thus, the surplus from his old match over continuing unemployment at that match quality must still be negative. Note that this is not true for workers who separate due to bargaining disagreement, but these do not exist on the equilibrium path. By assumption, a new match occurs only if it pays to start production right away, rather than to just mothball it. Therefore, if the surplus it generates over separating and keeping the *new* match quality is positive, then, to be acceptable, the new match must pay the worker more than the recall option he already had in hand.

Formally, we guess and later verify that the functions U and $W - U$ (hence W) are increasing in ε . Consider a worker who decides in this period not to recall the old match ε (i.e., $W(p, \varepsilon) - U(p, \varepsilon) \leq 0$), searches for new vacancies, and draws a new match $\tilde{\varepsilon}$ that is acceptable (i.e., $W(p, \tilde{\varepsilon}) - U(p, \tilde{\varepsilon}) \geq 0$). Combining the two inequalities, $W(p, \tilde{\varepsilon}) - U(p, \tilde{\varepsilon}) \geq 0 \geq W(p, \varepsilon) - U(p, \varepsilon)$. As $W(p, \cdot) - U(p, \cdot)$ is increasing, this implies $\tilde{\varepsilon} \geq \varepsilon$. As $U(p, \cdot)$ is increasing, this in turn implies $U(p, \tilde{\varepsilon}) \geq U(p, \varepsilon)$. Putting everything together, when an unemployed worker holding a mothballed match of quality ε finds a new match $\tilde{\varepsilon}$ that would be acceptable even if he did not have a recall option, he will accept it anyway: $W(p, \tilde{\varepsilon}) \geq U(p, \varepsilon)$. To conclude: a searching worker will accept a new match independently of his value of recall, as encoded in ε .

Firms post vacancies until job market tightness θ equates the expected hiring cost to the expected surplus from an acceptable new match:

$$(5) \quad \kappa = q(\theta) \int I\{J(p, \tilde{\varepsilon}) \geq V(p, \tilde{\varepsilon})\} J(p, \tilde{\varepsilon}) dF(\tilde{\varepsilon}),$$

which shows that indeed $\theta = \theta(p)$ is uniquely determined as a function of p only. Although the current new vacancy is worth zero, the firm knows that it will gain the surplus $J(p, \tilde{\varepsilon})$ over it only if the new match draw $\tilde{\varepsilon}$ is good enough also to start production right away, $J(p, \tilde{\varepsilon}) > V(p, \tilde{\varepsilon}) > 0$, because new matches cannot be mothballed before production.

It follows that the probability $a(p, \varepsilon)$ that a worker who is unemployed accepts a new offer is independent in equilibrium of the value of the recall option ε he has in hand, and so depends only on the aggregate state. We can write it as

$$A(p) = \int I\{W(p, \tilde{\varepsilon}) \geq U(p, \tilde{\varepsilon})\} dF(\tilde{\varepsilon}).$$

Again, we stress that this “memoryless” property applies only on the equilibrium path because it relies on all unemployed workers holding a negative surplus from recalling their last match. Still, to calculate the outside options for wage bargaining, which we assumed to be temporary separation until the next productivity shock, we need to know the current match quality ε . These types of separations, however, are off the equilibrium path, and hence do not affect the pool of unemployed from which new vacancies draw. Crucially, these separations do not affect the free entry condition.

E. Nash Bargaining and Wages

We assumed that the outside options are the continuation values of separating until at least the next productivity shock hits. Examining the Bellman equations, these are precisely $U(p, \varepsilon)$ and $V(p, \varepsilon)$. Therefore, the Nash bargaining solution is

$$(6) \quad w(p, \varepsilon) = \arg \max_w [W(p, \varepsilon) - U(p, \varepsilon)]^\beta [J(p, \varepsilon) - V(p, \varepsilon)]^{1-\beta}.$$

Taking a FOC yields¹⁸

$$(7) \quad \beta J(p, \varepsilon) = (1 - \beta) [W(p, \varepsilon) - U(p, \varepsilon)].$$

Using the Bellman equations and (7), and after much algebra, we can solve for the wage:

$$(8) \quad w(p, \varepsilon) = (1 - \beta)b + (1 - \beta) [s^*(p, \varepsilon) \phi(\theta(p)) \Omega(p, \varepsilon) - c(s^*(p, \varepsilon))] + \beta p \varepsilon \\ + \beta s^*(p, \varepsilon) \phi(\theta(p)) a(p, \varepsilon) V(p, \varepsilon) \\ + \lambda_\varepsilon \int [\beta V(p, \varepsilon') - (1 - \beta) U(p, \varepsilon')] [dG(\varepsilon' | \varepsilon) - dH(\varepsilon' | \varepsilon)].$$

¹⁸Shimer (2006) points out that in the case of costly search *on the job* the Nash problem (6) may not be concave, so the necessary FOC that yields the standard linear sharing rule (7) may not be sufficient. Intuitively, the firm may want to offer a higher wage than that implied by (7) in order to discourage job search by its employees, gaining on net due to retention, at the expense of future employers. This issue does not arise in our context, and (7) is sufficient for (6), because search effort occurs only *off the job*. The firm cannot commit to a future wage conditional on a recall in order to influence the worker's current incentives to search off the job while waiting for that recall. Once the recall occurs, bygones are bygones, wages are renegotiated ex post, and on-the-job search is ruled out by assumption.

The worker is paid the flow value of being unemployed, which includes leisure b and the surplus from searching optimally for a new match, plus his bargaining share β of flow output minus this opportunity cost and (second line) of the potential loss to the firm of the recall value, should the worker indeed find a new viable match. Note that the employed worker would search if temporarily separated because of bargaining disagreement (off the equilibrium path), in which case his propensity to accept a new job *would* depend on the recall option ε . Finally, in the third line, the wage contains a term that captures the differential evolution of match quality on and off the job. Suppose $G(\cdot|\varepsilon) \succ_{FSD} H(\cdot|\varepsilon)$, i.e., starting from ε , match quality improves on the job relative to off the job, for example because of match-specific skill depreciation during unemployment. Then the last wage component is positive if and only if $\beta V(p, \varepsilon') - (1 - \beta) U(p, \varepsilon')$ is increasing in ε' , i.e., if (what the worker can appropriate of) the firm's recall option is more sensitive to match-specific shocks than (what the firm can appropriate of) the worker's recall option.

F. Equilibrium

The equilibrium of the model is described by functions J , V , W , U , w , of ε and p , and a function θ of p , which solve (1), (2), (3), (4), (5), and the sharing rule (7) or, equivalently, the wage equation (8). It is straightforward to solve this system of functional equations exactly through any nonlinear iteration algorithm, after discretizing the support of ε and p . We exploit this tractability to explore the quantitative properties of the model.

Before doing that, we show that our model nests the standard search-and-matching framework as a special case without recall and worker search effort. Equations (1), (3), (5), and (7) are unaffected. Eliminating the recall option ($H = 0$), free entry $V(p, \varepsilon) = 0$ for all (p, ε) replaces (4), and $U(p, \varepsilon) = U(p)$. Appropriately modifying (2), the NB wage solution (8) reduces to

$$w(p, \varepsilon) = (1 - \beta)b + (1 - \beta) \left[s^*(p, \varepsilon) \phi(\theta(p)) \Omega(p, \varepsilon) - c(s^*(p, \varepsilon)) \right] + \beta p \varepsilon.$$

Eliminating worker search effort ($c(s) = 0$, and normalizing $s = 1$), we recover two well-known cases. In steady state, we obtain $w(\varepsilon) = \beta \varepsilon + (1 - \beta)b + \beta \theta \kappa$, which is the standard wage function of the classic models of Pissarides (1985) with initial, but fixed match heterogeneity F and no further idiosyncratic shocks, and of Mortensen and Pissarides (1994), where all new matches are the same but are then subject to idiosyncratic shocks. With aggregate but no idiosyncratic shocks, the wage is

$$w(p, \varepsilon) = \beta p \varepsilon + (1 - \beta)b + \beta \lambda_p \int \max \langle J(p', \varepsilon), 0 \rangle dP(p'|p) + \beta \theta(p) \kappa,$$

which corresponds to Shimer's (2005) stochastic version of Pissarides (2000).

VI. Quantitative Analysis

A. Calibration

We calibrate the model in steady state and then explore its business-cycle properties. A unit time interval in the model is set equal to a week. We simulate the model's steady-state equilibrium to generate a weekly panel. After discarding the observations in a “burn-in” period, we resample the data every four weeks and compute the cross-sectional model-based statistics. We do so to be consistent both with the structure of SIPP interviews and with the continuous-time setup of the model economy. A similar simulation procedure is used for the business-cycle analysis, whose results are described in Section VIB. The computational methodology is presented in the online Appendix.

We begin with normalizations, externally calibrated parameters, and functional forms. The discount rate is set to $r = 0.1$ percent, which roughly corresponds to 5 percent at annual frequency. We normalize to 1 the unconditional means of idiosyncratic and aggregate productivity, ε and p . In steady state, the latter takes the constant value $p = 1$. The contact rate of unemployed workers with open vacancies, per unit of time spent searching, derives from a standard Cobb-Douglas matching function: $\theta q(\theta) = \mu \theta^\alpha$, where job market tightness θ is the ratio between open vacancies and aggregate search effort of the unemployed, and μ is a matching scale parameter. In steady state, the scale of μ and $\bar{\theta}$ are not separately identified, so we normalize $\bar{\theta} = 1$. We set $\alpha = 0.5$, a standard number in the literature, and the worker bargaining share $\beta = 1 - \alpha$, a tradition that originates in the Hosios condition for constrained efficiency, although this condition need not apply to our economy. We set $\lambda_\varepsilon = 3/13$, so that idiosyncratic shocks to $\varepsilon > 0$ arrive on average every 13/3 weeks (i.e., one month), and $\lambda_p = 1/13$, so that aggregate shocks to p arrive on average once per quarter.

Next, we move on to the parameters that we calibrate internally. Conditional on the arrival of an idiosyncratic shock, the match experiences exogenous destruction with probability δ : match productivity transits from any state $\varepsilon > 0$ to the lowest state $\varepsilon' = 0$, which is absorbing, making any future recall impossible. The remainder of the EU transitions are endogenous separations. With probability $1 - \delta$, $\log \varepsilon$ experiences an innovation drawn from an AR(1) process with parameters ρ_ε and σ_ε . This compound process determines G . After separation, match quality evolves according to the same stochastic law of motion with no skill depreciation: $H = G$. We constrain search effort s in $[0, 1]$ and interpret it either as the fraction of time spent for job search or the flow probability of search by the unemployed worker. The search effort cost function is quadratic: $c(s) = c_0 s^2/2$.

We calibrate seven parameters (ρ_ε , σ_ε , δ , μ , c_0 , κ , and b) by minimizing the log unweighted distance of a vector of nine moments generated by the steady-state equilibrium from their empirical counterparts. Consistent with the model, where workers always participate in the labor force, these moments are computed from completed unemployment spells EUE , hence excluding entrants. We start with seven transition moments. The first two are standard in the literature, so to facilitate comparison with it, we draw them from the matched records of the monthly CPS 1990–2014. The total EU separation probability is 1.4 percent per month, and the total UE job-finding

probability is 27.7 percent per month. The remaining five moments are computed from the SIPP 1996–2013. The recall share of hires is 46.4 percent, which implies a new-job-finding probability of 14.85 percent per month. The hazard rate of exit from unemployment to employment is 35 percent after one month and 25 percent after six months. The analogous hazard rates of exit to just recall are, respectively, 20 percent and 10 percent.

Our choice of these empirical targets is motivated by the following considerations. Job-finding and separation probabilities are at the core of the model; they directly impact the unemployment rate and the probability of recall. The four moments on duration dependence are informative about the selection effect by match quality, which is, in our model, the source of recall. In the data, unemployment spells exhibit negative duration dependence mostly when the spell ends with recall, and we aim to replicate this property.

The eighth target is aggregate search effort, or equivalently the share of the unemployed who search full time. We take it to be the average share of PS workers in the unemployment pool (excluding entrants), which is 80 percent in the CPS over the period 1990–2014. Although every unemployed worker in the model will spend some fraction of his time searching, this fraction will be increasing in the distance of ε to the recall threshold; hence, search effort and probability of recall will be negatively correlated across workers. In the data, the coarse categorization PS/TL satisfies this negative correlation, so we view our choice as a reasonable approximation.

A critical parameter to calibrate in this literature is the flow value of leisure b . Brügemann and Moscarini (2010) work with steady-state equilibrium equations that apply to a large class of search models. They study the comparative statics response of the job finding probability to changes in aggregate productivity p and show that this response, an upper bound to the volatility in the stochastic simulation of the same model, depends directly on the ratio b/p . Their findings generalize the insight from Hagedorn and Manovskii's (2008) calibration of Shimer's (2005) specific model. Here, a similar argument, omitted and available upon request, shows that the source of amplification is the ratio between the value of leisure b net of average search costs when unemployed and average labor productivity corrected by match selection through endogenous separations. We follow Hall and Milgrom (2008) and target this "replacement ratio" to be 0.71.

The parameters of the idiosyncratic shock process, ρ_ε , σ_ε , and δ , are the only drivers in the model of job separation and recall and of unemployment duration dependence, which we showed in the data to be mostly about declining recall chances. More persistent (higher ρ_ε) and smaller (lower σ_ε) innovations to match-specific productivity raise the equilibrium separation cutoff, thus, the probability of separation, and reduce the probability of recall. The latter effect is stronger the longer a worker has been unemployed (by selection), which shows up as negative duration dependence of recall. A higher chance δ of exogenous, irrevocable separation has similar effects on the chances of separation and recall but no impact on unemployment negative duration dependence. Therefore, observed duration dependence separately identifies the sources of exogenous and endogenous separations and recalls.

Table 6 summarizes our best calibration. The implied value of leisure (b) is 0.9, while the average search cost paid is 0.11. Thus, the net benefit is 0.79 and the

TABLE 6—PARAMETER VALUES: WEEKLY CALIBRATION

	Description	Value
r	Discount rate	0.001
μ	Matching scale parameter	0.067
b	Flow value of unemployment	0.9
κ	Vacancy posting cost	0.722
c_0	Search cost scale	0.29
β	Worker bargaining share	0.5
λ_ε	Arrival rate of idiosyncratic shock	3/13
α	Matching function elasticity	0.5
δ	Exogenous job destruction rate	0.0005
λ_p	Arrival rate of aggregate shock	1/13
ρ_ε	Persistence of idiosyncratic shock	0.97
ρ_p	Persistence of aggregate shock	0.97
σ_ε	Standard deviation of idiosyncratic shock	0.035
σ_p	Standard deviation of aggregate shock	0.008
—	Mean output level	1

replacement ratio to average productivity of active matches (1.06 in our calibration) is 0.75, slightly above the 0.71 target. The flow surplus from employment, however, remains substantial.

We can now examine the implications of our calibration for a few untargeted moments. An exogenous job destruction probability δ equal to 0.05 percent per week implies that the share of workers with no recall option at the time of job separation is about 17 percent in the steady state. To gauge its plausibility, we observe that, when an establishment closes, its employees presumably lose any recall option. According to the Business Employment Dynamics assembled by the BLS, each quarter, establishment closings result in about 1.5 million job losses. This is about 25 percent of total *EU* separations in the CPS. Some of the workers affected by closings may not be part of the *EU* inflow because they are reassigned to a different establishment owned by the same firm, or find another job right away, or drop out of the labor force.

Our calibrated idiosyncratic productivity process (in logs) has persistence 0.97 monthly, meaning 0.7 annually, and a standard deviation of innovations equal to 3.5 percent monthly, 8 percent annual, and 14 percent ergodic. The employer-level (log) TFP estimated by Foster, Haltiwanger, and Syverson (2008) has exactly the same persistence, 0.7 annual, and higher volatility, 21 percent over five years. Given possible sources of measurement error, we find this to be in the ballpark; our model does not require implausibly volatile or transient measures of productivity.

The implied average contact rate of new vacancies per unit of search effort, namely of a full-time job searcher ($s = 1$), which is $\bar{\theta}\bar{q} = \bar{q}$ given the normalization $\bar{\theta} = 1$, equals 6.7 percent per week. Combined with an 80 percent probability of accepting a new match, in the calibrated model the probability for a vacancy of hiring a *new* worker from unemployment is about 5 percent per week, or 20 percent per month. This is only about one-quarter of the 80 percent average vacancy-filling probability measured in the *Job Openings and Labor Turnover Survey* (JOLTS).¹⁹

¹⁹<https://www.bls.gov/jlt/>.

Three differences in definitions can help to explain this discrepancy. First, our notion of a vacancy does not coincide with JOLTS's: the model's cyclical dynamics do not depend on the scale in which we measure vacancies, and for that reason we normalized it to equal steady-state unemployment ($\bar{\theta} = 1$). If we rescale vacancies in the model to match $\bar{\theta} = 0.5$, which is roughly the average ratio between the number of vacancies from JOLTS and the number of unemployed workers in the CPS during 2001–2016, then the implied job-filling rate by new workers in the model doubles to 40 percent per month. Second, the model's predictions concern the rate at which workers join *new* firms, while hires in JOLTS include recalls, about one-third of all hires from non-employment according to our evidence in online Appendix Table A.4. This correction reduces the JOLTS filling rate by one-third, to 53 percent, much closer to our 40 percent. Finally, hires in JOLTS also include those from nonparticipation and from other firms, which are outside of the scope of the model. Job-to-job transitions alone account for about one-half of all hires, based on evidence from the CPS and the SIPP. If we exclude those hires from the numerator of the JOLTS job-filling rate, presumably we should also discount a share of open vacancies in the denominator, by an amount which is difficult to assess without a model.

Tables 7 and 8 report the fit of the model. Quantitatively, this simple calibration with a parsimonious idiosyncratic process does a remarkable job at fitting both targeted and nontargeted empirical moments.

B. Cyclical Properties of the Model

We now examine the cyclical properties of the model's equilibrium. To calibrate the aggregate productivity process p , we assume that, conditional on an arrival at Poisson rate $\lambda_p = 1/13$, it follows an AR(1) process in logs. We calibrate the innovations' serial correlation at 0.97 and standard deviation at 0.008 in order to replicate the cyclical properties of Average Labor Productivity (ALP), including the "cleansing" effect of recessions. To measure ALP, we follow Shimer (2005) and use "output per job in the nonfarm business sector" from the BLS (series PRS85006163) over the same sample period (1990–2014) as in the case of the CPS transition rate series discussed above. The cyclical component is measured by its logged and HP-filtered series with smoothing parameter of 10^5 . Rather than calibrating the aggregate driving process to a specific series, such as the Solow residual or identified monetary policy shocks, we take this agnostic view, because our model features a single aggregate shock, while in the data there are several. To facilitate comparison with the literature, we target ALP, but in our model this is an endogenous object. Our focus is on comovement and amplification of labor market variables, not on the origin of aggregate economic fluctuations. We measure comovement, both in the data and in the model, with the semi-elasticity (regression coefficient) of each relevant variable on unemployment rather than the unconditional correlation between the two, which is contaminated in the data by additional shocks.

The main goal of our modeling exercise is to understand the impact of the recall option on aggregate labor market fluctuations. To this end, we also present results from versions of the model where we remove search effort and/or recall. We label

TABLE 7—FIRST MOMENTS IN THE MODEL STEADY-STATE EQUILIBRIUM AND EMPIRICAL TARGETS

	Job-finding prob.	Separation prob.	Recall rate	Search prob.	Replacement ratio
Model	0.291	0.014	0.499	0.791	0.75
Data	0.277	0.014	0.464	0.800	0.71

Note: Sample period for CPS-based measures is 1990–2014.

TABLE 8—MEAN UNEMPLOYMENT DURATION AND HAZARD RATES IN THE MODEL STEADY-STATE EQUILIBRIUM

	Overall hires	Recalls	New hires
Mean duration (months)	3.41	2.80	4.01
Hazard rate at month 1	0.346	0.204	0.142
2	0.318	0.169	0.148
3	0.290	0.139	0.150
4	0.274	0.124	0.151
5	0.264	0.110	0.154
6	0.257	0.098	0.159

Notes: The empirical counterparts are in Table 4 and Figure 1. Note that Figure 1 includes those who drop out of the labor force, whereas the calibration targets are hazard rates of the *EUE* sample.

“MP” the model without recall, essentially identical to Mortensen and Pissarides (1994) with the only difference that we also allow for an interesting acceptance margin, as in Pissarides (1985), while in Mortensen and Pissarides (1994) all new matches are acceptable.²⁰ We recalibrate the two versions of the MP model (with and without search effort) in steady state by targeting the average job-finding probability and separation probability. These two moments alone are insufficient to identify all parameters. For example, the values of both δ and σ_ε govern *EU* separations. In our benchmark model in which shocks to ε keep hitting even after separation, these two parameters also determine the frequency and duration dependence of recall. In the MP model, we keep the calibration of our benchmark model and only modify three parameters, which speak directly to these two moments: σ_ε , the scale parameter of the matching function μ , and the vacancy posting cost κ . We also reset the value of leisure b to maintain our target replacement ratio. When introducing search effort in the MP model, we recalibrate the scale of its cost function to target the share of PS with average search effort, as in our recall model. We report in the online Appendix the resulting values of the parameters for these calibrated models without recall. Finally, we compute their stochastic equilibrium in response to the same sequence of aggregate shocks as the benchmark model with recall and search effort.

We log and HP-filter (with parameter 10^5) all time series, empirical and model-generated, sampled quarterly. Tables 9 and 10 present the standard deviations of the

²⁰This exercise is of independent interest, as the first quantitative exploration of business cycles in a canonical search-and-matching model simultaneously featuring endogenous rates of match contact, acceptance, and separation. See Fujita and Ramey (2012) for the cyclical properties of various versions of the Mortensen and Pissarides (1994) model, where new matches are always accepted.

TABLE 9—STANDARD DEVIATIONS IN THE MODEL STOCHASTIC EQUILIBRIUM AND EMPIRICAL TARGETS

Model	Search cost	ALP	Separation prob.	Job-finding prob.	Measured tightness	Recall rate
Recall	Yes	0.016	0.152	0.088	0.169	0.068
	No	0.017	0.095	0.040	0.066	0.034
MP	Yes	0.017	0.087	0.106	0.144	—
	No	0.018	0.061	0.041	0.072	—
Data		0.016	0.103	0.145	0.350	0.082

Note: Measured tightness equals the ratio between vacancies and unemployment.

series and their semi-elasticities (regression coefficients) with respect to the unemployment rate, in the Recall models, MP models, and the empirical data.

Both the recall option and search effort amplify countercyclical fluctuations in the probability of endogenous separation. In our benchmark model with recall and search effort, the volatility of the unemployment rate (not shown) is 0.199, comparable to its empirical counterpart. This happens, however, in part for the wrong reason: the separation probability into unemployment is 1.5 times as volatile in the model (first row) as in the data (last row), while close to the opposite is true for the overall job-finding probability. Given that our calibration does not target aggregate second moments, this result is not surprising. On the other hand, the MP model without recall and search effort (fourth row), which is the natural term of comparison, underestimates by much more the volatility of *both* job-finding and separation probabilities, and therefore the volatility of the unemployment rate. As we know from Shimer (2005), this MP model does even worse on both dimensions.²¹ We conclude that adding to the MP model the recall option, in a way that is consistent with our empirical evidence, does not fully resolve the unemployment volatility puzzle of Shimer (2005), but goes in the right direction.

The intermediate models, in the second and third rows of Table 9, reveal that both recall and search effort amplify the volatility of the separation probability, while they have opposite effects on the volatility of the job-finding probability. Removing search effort from the recall model, in the second row of Table 9, improves the volatility of the separation probability but rolls back any gains in the volatility of the job-finding probability, which is the main focus of the literature. The recall rate as well becomes too stable.

Removing the recall option while leaving search effort (third row of Table 9) reduces the volatility of the separation probability and raises that of the job-finding probability. This MP model without recall but *with* search effort appears to do best and appears to negate the importance of recall and our entire exercise. But this version of the MP model is simply a term of comparison with our recall model, useful only to inspect the mechanism. Nothing in the logic of the MP model suggests how to calibrate the cost of search effort, which is critical to this intermediate result. We simply matched the share of unemployed workers who are on PS, as in the recall model, for the sake of comparison. But the TL/PS distinction does not really

²¹ For the separation probability, this is by construction, given that it is assumed to be constant in his model.

TABLE 10—ELASTICITY WITH RESPECT TO UNEMPLOYMENT

Model	Search cost	Separation prob.	Job-finding prob.	Vacancies	Recall rate
Recall	Yes	0.690	−0.404	0.174	0.252
	No	0.820	−0.294	0.458	0.096
MP	Yes	0.412	−0.663	0.122	—
	No	0.622	−0.501	0.180	—
Data		0.493	−0.756	−0.854	0.222

belong in the MP model, and even the PS share alone provides weak identification for the search technology (for example, our choice to model it as a quadratic cost). Conversely, the TL/PS distinction is natural in the recall model, given the strong empirical correlation between TL and recall (and PS and no recall). In both the SIPP and the CPS, the definition of TL explicitly ties an expectation of recall to the measurement of search effort. Finally, the cyclical volatility of the recall rate (last column of Table 9) provides additional empirical discipline to evaluate the specification and calibration of the search technology.

In our model, “true” labor market tightness is the ratio of vacancies to aggregate search effort. In the data, we can only observe (the ratio between) vacancies and unemployment. As Shimer (2005) points out, this vacancy/unemployment ratio, “measured” tightness, is roughly 20 times more volatile than ALP (0.35 versus 0.016 in our data). We can replicate measured tightness in the model. Its volatility is larger, roughly one-half of the empirical counterpart, in the recall model (0.169) than in the MP model with search effort (0.144).

We now turn to comovement in Table 10. Our benchmark recall model with search effort and its polar opposite MP model without search effort perform similarly. As is clear from the intermediate models, recall and search effort have countervailing effects. The entries in the vacancies column measure the slope of the empirical Beveridge curve. Along this dimension, all models perform poorly, with a wrong, positive sign. The recall model without search effort performs worst, although introducing search effort brings the elasticity down to a level comparable to the two versions of the MP model. This poor fit of the Beveridge curve is a well-known implication of countercyclical separations. The literature has shown that on-the-job search and associated job-to-job transitions overcome this problem (Fujita and Ramey 2012). The same is likely to apply to the model with recall. Finally, our benchmark model replicates well the mildly countercyclical behavior of the recall rate, which acts as a stabilizer of total hires. This effect is due mostly to search effort, which falls in a recession, making recall a more likely outcome of unemployment.

C. Discussion

We now interpret our quantitative results. Table 11 provides additional informative moments. First, why does the recall option amplify the volatility of separations? When deciding whether to separate, a firm is concerned that the mothballed worker may find another job and become unavailable for recall. This concern is stronger in expansions when both search effort and the probability of contacting new vacancies

TABLE 11—CYCLICALITY OF “NEW HIRES” JOB-FINDING PROBABILITY

Model	Search cost	Job-finding prob. (new hires)	Search prob.	Acceptance prob.	Tightness
<i>Standard deviation</i>					
Recall	Yes	0.137	0.072	0.034	0.098
	No	0.048	—	0.024	0.066
<i>Elasticity with respect to unemployment</i>					
Recall	Yes	−0.659	−0.349	−0.135	−0.476
	No	−0.377	—	−0.134	−0.542

rise, and hence, a firm is more reluctant to separate and hoards even more labor. Conversely, in recessions, a firm is more willing to mothball its unproductive workers, who have nowhere to go. Because the separation cutoff is also the cutoff for the acceptance of new matches, by the same logic, the probability of accepting a new job offer also becomes more cyclically volatile. Intuitively, the ability of the worker to search for other jobs while waiting for a recall and to void the recall option raises the surplus from staying together. This force is stronger in expansions, and further encourages firms to post vacancies, as their offers are more likely to be accepted, so (true) tightness also moves more. For all these reasons, the recall option makes the probability of finding *new* jobs more cyclically volatile. The *total* job-finding probability, however, contains many recalls, which are much more stable, so it is slightly less volatile in the model with recall. Note the interesting tension between model and data: accounting for recalls improves amplification in the model, but also makes its task harder by raising the target volatility of the new-job-finding probability estimated from the data.

Recall is stable in the model as a result of three, partially opposing forces. First, a positive aggregate shock encourages production: a whole set of unemployed workers is recalled on impact when the economy improves and the separation/acceptance cutoff falls; as the expansion unfolds, some previously unlikely recalls become plausible for idiosyncratic reasons. Second, the quality of idle matches is strongly countercyclical. We compute at each point in time the ratio between the average “shadow” productivity of the unemployed, based on the still-evolving quality of the last match, and the actual productivity of the marginal match at the cutoff. The regression coefficient of this ratio with respect to the unemployment rate is positive at 0.015. The intuition behind this cyclical selection is simple. The probability δ with which match quality drops permanently to its lowest (zero) value, voiding any recall option, is acyclical, while the endogenous separation probability into the unemployment pool, from which recall is possible, is countercyclical. So, in a recession, a larger fraction of unemployed workers are recallable. Finally, the incentives to search for new jobs are procyclical, while the idiosyncratic shocks leading to recall are acyclical. Therefore, in a recession, separated workers spend much longer being unemployed and are more likely to be available for a recall.

Turning to search effort, it is clearly procyclical: in expansions there are more jobs available, and the return from working is higher. As workers search harder for new vacancies, firms post more of them, so tightness responds strongly to aggregate shocks. Therefore, endogenous search effort amplifies the response of the job-finding probability, both directly and through its effect on vacancy postings. Search effort

also raises the volatility of separations, strongly interacting with recall. Without a search effort margin, in an expansion, a firm is less concerned about mothballing a worker who cannot increase his search effort to take advantage of good aggregate conditions; thus, separations decline by less. The opposite is true in a recession.

Whether job search effort by the unemployed is pro- or counter-cyclical (here meaning negatively or positively correlated with the unemployment rate) is an important issue of difficult empirical resolution. The main stylized fact traditionally cited in this regard is the number of job search methods used by a typical unemployed worker, which increases with the unemployment rate (e.g., Shimer 2004, and an older literature reviewed in Moscarini 2001); yet, different search methods may differ in efficacy and in intensity of use. Recently, Mukoyama, Patterson, and Şahin (forthcoming) provided direct evidence, based in part on the *American Time Use Survey* (ATUS),²² that unemployed workers search for jobs harder at times of high unemployment. The short-time span of the ATUS requires an imputation of time spent on job search, based on observations available only during the single (and special) 2008–2009 recession. On the extensive margin, after recessions non-employment shifts away from nonparticipants to the unemployed, who search more on average; within unemployment, the composition changes toward the long-term unemployed, who are in the ATUS the most likely to spend longer per day searching for a job when faced with a loose labor market, or with looming expiration of their unemployment benefits. Our model abstracts from the participation margin, so the first composition effect is not directly relevant. On the intensive margin, the time that the average unemployed worker reports searching for employment in the ATUS is clearly countercyclical, off a suspiciously low average of less than one hour per day. But within short-term unemployment, which is where recall has bite, the average daily time spent on job search is procyclical (Gomme and Lkhagvasuren 2015) as in our model.

Finally, in our model, aggregate search effort effectively makes matching efficiency positively correlated with job market tightness. Borowczyk-Martins, Jolivet, and Postel-Vinay (2013) point out that, under these circumstances, an OLS regression of the job-finding rate on job market tightness is bound to overestimate the matching function elasticity. They propose and implement a GMM procedure to correct for this bias and indeed find strong evidence that it is positive.

To conclude, all four models fail to replicate the negative correlation between unemployment and vacancies that constitutes the empirical Beveridge curve. This is a well-known feature of search models like Mortensen and Pissarides (1994), where endogenous separations increase in recessions the pool of unemployed workers who are available for a fresh rematch, spurring vacancy postings. This effect is never strong enough, however, to make tightness countercyclical. Models of purely exogenous separations like Shimer (2005) do well with the Beveridge curve but miss by construction the remarkable countercyclical volatility of the *EU* separation probability. As explained, recall amplifies fluctuations in the separation probability, so per se it makes the problem even worse. Search effort, however, compensates, because it is naturally procyclical, so effective aggregate search effort is not as countercyclical

²²<https://www.bls.gov/tus/>.

as unemployment. The argument that the MP model is mostly about job creation, justifying the simplifying assumption of a constant separation rate, does not apply to our version with recall, where temporary separations directly interfere with new hires. Future research will need to address this interesting tension.

D. Alternative Calibration

One might argue that an alternative strategy to examine the impact of recall is to change the calibration rather than the model. Based on the idea that recalls are not mediated by a matching function, we can ignore them, and the separations that precede them, altogether when estimating transition rates. We calibrate the canonical search-and-matching MP model by matching the average probability that an unemployed worker finds a *new* job, which is only 15 percent per month as opposed to 27.7 percent, and that an employed worker is permanently separated into unemployment, 0.9 percent per month as opposed to 1.4 percent.

This different steady-state calibration hardly affects the model's ability to amplify aggregate shocks. The implied elasticity of measured job market tightness to unemployment changes only by a tiny amount, from 3.63 to 3.70. These values double the 1.78 obtained by Shimer (2005), because the value of b/p is nearly doubled in our calibration, but are still much too low, in line with the low volatility of the job-finding probability in our stochastic MP model with no search effort. The reason for this invariance result can be traced to the properties of the standard model. As the literature made clear, the amplification of aggregate shocks in the MP model can be bounded above by performing comparative statics on its steady-state equilibrium. Shimer (2005) calculates the elasticity of the steady-state job-finding probability $\phi(\theta)$ with respect to aggregate labor productivity p to be, in our notation:

$$\frac{1}{1 - b/p} \cdot \frac{r + \delta + \beta\phi(\theta)}{(r + \delta)(1 - \alpha) + \beta\phi(\theta)}.$$

The second fraction is always close to 1 as long as the bargaining share β is non-negligible, because the empirical job-finding probability $\phi(\theta)$ is always, even in our alternative calibration, an order of magnitude larger than $r + \delta$. Simply put, in the US economy, it is much easier to find a job than to lose one, and in fact easy enough to make discounting irrelevant.

We conclude that, in order to make quantitative progress in our understanding of cyclical unemployment, our new evidence on recall must be incorporated in the model and not only in the calibration targets. Recall is not only a matter of measurement but also a matter of incentives, for firms to mothball their workers and for workers to search for a new jobs, both of which change over the business cycle.

VII. Conclusions

In this paper, we document that US workers who separate from their jobs have a surprisingly high probability of going back to the same employer and that the share of such recalls out of all hires from unemployment is countercyclical. Recalls involve mostly workers on temporary layoffs, but also many permanently separated workers.

Recall is more likely the longer the worker had spent at that employer before separation and is associated with dramatically different outcomes in terms of unemployment duration (both the level and shape of the exit hazard) and post-reemployment attachment. Recalls are relatively stable over the business cycle, so that the rate of exit from unemployment to new jobs is even more volatile than previously estimated. A relatively modest modification to the canonical Mortensen and Pissarides (1994) model of unemployment, embedded in a business-cycle framework, captures well these empirical patterns through selection of workers to be recalled. Recall, through its effect on expectations and job search effort, amplifies the business cycle volatility of the average job-finding and separation probabilities.

We believe that these findings cast our knowledge of the aggregate labor market under a different light. In future work, we will explore the implications of our empirical findings for the importance of firm- and occupation-specific human capital. We will also revisit more deeply, under the lens of our new stochastic search-and-matching model with recall, classic questions in this field, such as the unobserved heterogeneity between short- and long-term unemployment, and the implications of establishment closings on earnings prospects of the displaced workers who lose the recall option.

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